

Master's Thesis

Modelling Traffic Incidents Impacts and Prediction of
Incident driven Congestion Propagation in a Large
Scale Road Network

Juyeong Lee

Department of Industrial Engineering
(Industrial Engineering)

Ulsan National Institute of Science and Technology

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Juyeong Lee

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Approved by



Advisor

Sungil Kim

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Juyeong Lee

This certifies that the thesis/dissertation of Juyeong Lee is approved.

07.09.2021

Signature



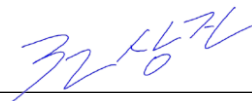
Advisor: Sungil Kim

Signature



Sungahn Ko: Thesis Committee member #1

Signature



Sang Jin Kweon : Thesis Committee member #2

Abstract

As cities around the globe are becoming more urbanized, traffic congestion has been causing numerous problems. Especially, with growing number of vehicles, the importance of modelling non-recurrent congestion caused by traffic incident has become greater. However, there are many cases when the information regarding accurate incident occurrence, recovery time, and various traffic state indicators such as speed, flow, density are absent. Therefore, this study proposes novel methods of modelling non-recurrent congestion under the two big themes with limited data sources.

First topic is quantifying the traffic non-recurrent incident impact and identifying the influential factors. This topic has become important for traffic agent to deal with incident impact in timely and effective way. There have been many attempts to analyze the problem. This study proposes a novel method of measuring not only duration time but also congestion propagation and speed drop ratio caused by nonrecurrent traffic incident, which measures the impact in temporal, spatial, and speed perspective using average speed data generated from real user trajectory. Furthermore, to prioritize the importance of variables, LIME (Local Interpretable Model-agnostic Explanations) is applied to the ensemble model that combines three most accurate models out of nine machine learning models we used in this study. Maintaining the uniqueness of individual incident case, we use LIME to obtain variable importance for individual incident case and average the them in order to generalize the the result.

Second topic is predicting non-recurrent congestion propagation patterns in the near future. This topic is especially carried based on the GPS navigation service provider perspective because it is crucial for GPS navigation systems to provide drivers with route recommendations for bypassing congested roads in real time. However, it is difficult to model non-recurrent congestion propagation based on the non-stationarity of propagation patterns and lack of sufficient relevant non-recurrent traffic data corresponding to a given traffic network. These limitations make it difficult to develop practical methods that can be applied to real traffic incidents in an online manner. To overcome this difficulty, we propose a novel online learning method using pattern matching techniques to predict the propagation patterns of non-recurrent traffic congestion. The proposed method assumes that current non-recurrent propagation patterns are similar to historical recurrent patterns.

We validated the both proposed methods using real user trajectory data provided from NAVER Navigation Corp and incident data obtained from Korea National Police Agency.

We believe that this study not only provides traffic operators with an insight on traffic incident management, but also provides navigation corporation with new solution for real-time route exploration during traffic incident situation.

Contents

I	Introduction	1
II	Traffic Congestion in Seoul	2
III	Definition of Terminologies	4
IV	Prioritizing Influential Factors for Traffic Incidents Impacts in a Large Scale Road Network	7
	4.1 Introduction	7
	4.2 Related Works	8
	4.3 Methodology	10
	4.4 Case Study: Incident on Gangbyeonbuk-ro	14
	4.5 Real Data Example: Traffic Incidents in Seoul	16
	4.6 Prioritizing Influential Factors for Traffic Incidents Impacts	18
V	A Propagation Prediction Method for Non-recurrent Traffic Congestion	21
	5.1 Introduction	21
	5.2 Related Works	22
	5.3 Methodology	23
	5.4 Experimental Results	28
VI	Conclusion	35
	6.1 Summary and Contributions	35

6.2	Limitation and Future research	35
	References	37
	Acknowledgements	43

List of Figures

1	The sources of traffic events, City of Seoul	2
2	The concepts of the proposed non-recurrent traffic congestion measures	11
3	A road network with a real incident in Seoul	14
4	Speed curve for each path	15
5	Incidents location in Seoul (from 02/09/2020 to 01/12/2020)	16
6	Methodology overview	24
7	Grouping search algorithm	25
8	Summary of the traffic incident data	29
9	Results of hyper-parameter optimization	31
10	Accuracy distribution	32

I Introduction

Traffic congestion has been an universal problem to the urban cities as the dramatic growth of population, and the corresponding increases of vehicles, infrastructure, and proliferation of delivery services have been contributing to the problem. Congestion can cause various social, environmental and economic problems by reducing the use of transportation infrastructure and increasing travel time, air pollution and fuel consumption.

Traffic congestion caused by a particular event, rather than by regular traffic congestion during rush hour is called non-recurrent traffic congestion [1]. This type of congestion happens due to unexpected events such as vehicle collision, road maintenance construction, or debris. Among these types of non-recurrent congestion, traffic incident is the most difficult to predict, but must be dealt with in a timely and effective way to alleviate the impact on road capacity and huge travel time loss and to avoid likelihood of a secondary crash.

To mitigate the adverse traffic incident impact, the traffic agents such as the National Transport Information Center (NTIC) or navigation platform must have prior knowledge of the aspects of the impact and their influential factors so that they can provide timely and reliable information about route guidance in real-time manner. Because traffic incidents occur spontaneously, a well-coordinated traffic incident management process can improve safety, traffic flow and clearance time [2]. The influence of the detected incident is usually unknown until it is completely cleared, which is why it is an essential operation for the traffic agents to predict in advance.

This in this paper, we suggest two main studies topics that attempt to model traffic incident.

- First, we propose a new methodology for measuring the impact of non-recurrent traffic congestion propagation and for prioritizing influential factors in the development of non-recurrent traffic congestion impact prediction models.
- Second, we propose novel method that predicts non-recurrent congestion propagation patterns in online base.

The rest of the dissertation is organized as follows. Section II describes non-recurrent congestion occurred in Seoul, Korea from September 2 to December 1, 2020. Section III will provides shared definition of terminologies used throughout the topics. Section IV and Section V cover the two themes that consist the dissertation work. Each Section covers introduction, related works, methodology, experiments and result. Finally, Section VI concludes with the contributions and limitations of the researches and suggests future research.

More details will be discussed in following sections.

II Traffic Congestion in Seoul

Traffic events indicate traffic condition caused by external factors, abnormal traffic flows, which inevitably results in traffic congestion. Seoul is the capital of the Republic of Korea and the world's 16th largest city. The total population in 2020 was about ten million and registered vehicles was about three million. With huge population ratio and high number of vehicles, the city undergoes various traffic events. Previous work has only described regular congestion phenomenon in Seoul [3, 4]. On the other

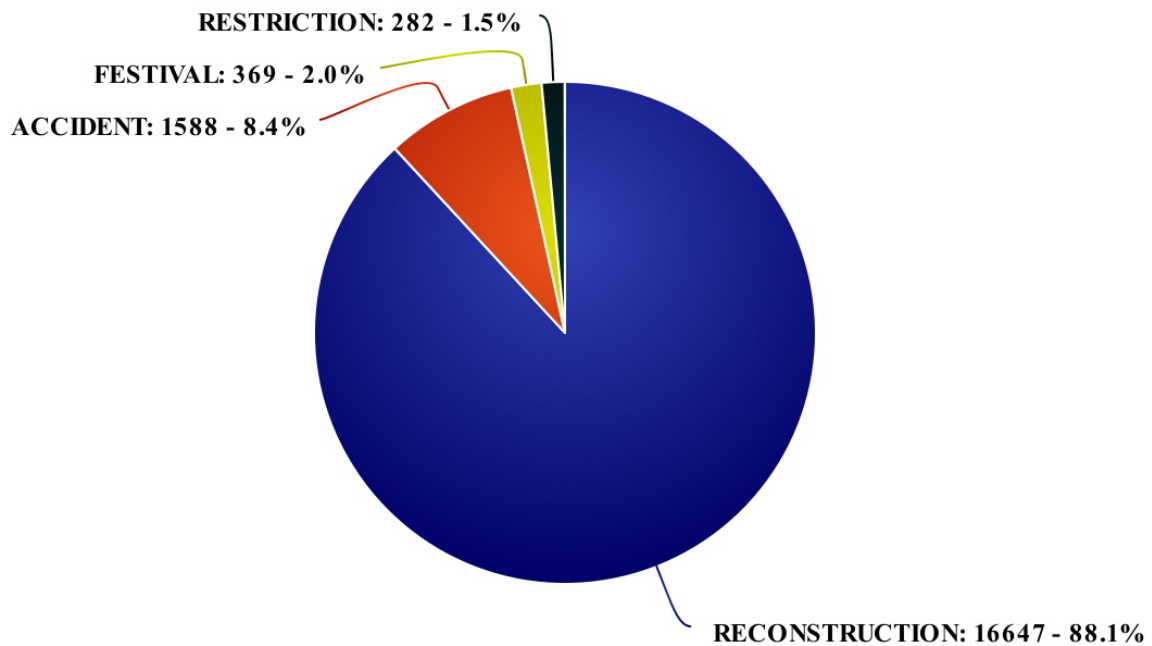


Figure 1: The sources of traffic events, City of Seoul

hand, we presents there are four main root causes of all 18,886 traffic events reported in city of Seoul during September 2 to December 1, 2020. incident data were obtained from Korea National Police Agency [5].

- **Reconstruction (88.1%):** Reconstruction holds the most portion of the entire events occurred in the city of Seoul. It causes physical changes in the highway environment. These changes may include a reduction or expansion in the number or width of travel lanes, lane diversions, reduction, or elimination of shoulders, and even temporary roadway closures. Constructions are going on for a long time which last for couples of weeks and even months, or years. During construction period, the roads are completely closed, which leads to traffic overflow on alternative roads.
- **Incident (8.4%):** Events such as vehicular crashes, breakdowns, and debris in travel lanes are the most common form of incidents. The traffic flow abruptly changes when an incident happens.
- **Festival (2.0%):** The city completely control the traffic flow where the festivals, mostly rallies, occur. Thereby, traffic demand in the vicinity of the events radically change from typical patterns. Special events occasionally cause dramatic surges in traffic demand in nearby roads, which

overwhelm the capacity that normally hold.

- Restriction (1.5%): Intermittent disruption of traffic flow by the City such as full control by cleaning road facilities, maintenance work, lamp structural improvement construction, and road pavement construction. This sort of 'Restriction' is normally released within a day or sometimes extended to few days.

For our study, among four major events in the city of Seoul, we are going to only focus on identifying critical relationships between the set of key factors of resulting incident impact. All events except for incidents, it is predictable how long the impacts of the event on restricted roads prolong. Roads are artificially controlled or restricted for estimated time period which construction, restriction, or festival is expected to last. Therefore, drivers are already aware of the traffic condition and can simply make a detour as if there are no roads nearby the events scene. On the other hand, an incident occurs unexpectedly. Drivers encounter an incident without any prior knowledge regarding the severity of an incident, which leads them to exposed to potential risks such as secondary incident, travel delay without any promises. Therefore, given the wide variety of causes and impacts of different kind of events, it is important for the traffic management system to understand how the impacts of an incident spread in which manner and to properly predict them based on influential factors contributing the impacts.

III Definition of Terminologies

Consider a road network, $G = (V, E)$, as a directed connected graph, where V is a set of nodes representing intersections and E is the set of road segments connecting the nodes. Let $v_i \in V$ denote a node and $e_{ij} = (v_i, v_j) \in E$ represent an edge. $v_i = \text{src}(e_{ij})$ means the source node of the edge e_{ij} , while $v_j = \text{dst}(e_{ij})$ is the destination node of the edge e_{ij} . In the graph, we define *in* and *out* operator such that the operator $\text{in} : V \rightarrow 2^E$ returns all the edges for which this node v is the destination and the operator $\text{out} : V \rightarrow 2^E$ returns all the edges whose source is node v . In addition, $|\cdot|$ is defined by the number of set elements. For example, $|\text{in}(v)|$ is the number of road segments incoming to the node, whereas, $|\text{out}(v)|$ is the number of road segments outgoing from the node.

Given a graph G , we assume that speed information of all road segment $e \in E$ is available for a pre-determined time point $t \in T$, which is denoted by $\{s(e, t) : E \times T \rightarrow \mathbb{R}^+\}$. In addition, we denote $s(e, T)$ by a time series of speed at road segment $e \in E$ during a predetermined time period T . Since $s(e, T)$ is a vector, we can easily define $\min(s(e, T))$, $\max(s(e, T))$, $\text{sum}(s(e, T))$, $\text{avg}(s(e, T))$, $\text{sd}(s(e, T))$, and so on. (sd stands for standard deviation.) For further analysis, we defined basic terminologies as follows.

Definition 1 (Exponential moving average): Let $s(e, t)$ be the speed on traffic incident road segment e at time t , then the exponential moving average $\bar{s}(e, t)_n$ can be defined recursively by

$$\bar{s}(e, t)_n = \alpha \bar{s}(e, t-1)_n + (1 - \alpha)s(e, t),$$

where n indicates the averaged time period [6] and a weight factor α plays a role to control the importance on recent information. While there are many possible choices for the weight factor, the most common choice is $\alpha = \frac{2}{n+1}$. The exponential moving average technique is typically used in time series data to smooth short-term variation and capture long-term trends or cycles by moving average.

Definition 2 (*k*-hop incoming(outgoing) neighbors): We define the *k*-nearest hops of the set of incoming edges feeding traffic into a target edge e_0 , called *k*-hop incoming neighbors, by

$$N_{in}^k(e_0) := \bigcup_{x \in N_{in}^{k-1}(e_0)} \text{in}(\text{src}(x)),$$

and, similarly, define the *k*-nearest hops of edges taking traffic away from a target edge e_0 via its out node, *k*-hop outgoing neighbors, by

$$N_{out}^k(e_0) := \bigcup_{x \in N_{out}^{k-1}(e_0)} \text{out}(\text{dst}(x)).$$

Definition 3 (Path): Given a road network $G = (V, E)$ with a set of nodes $V = \{v_1, \dots, v_n\}$ and a set of edges $E \subseteq V \times V$, a path in G is defined as finite sequence

$$P = (p_1 e_{p_1 p_2} e_{p_2 p_3} \dots e_{p_k p_{k+1}})$$

with $p_i \in V$ for all $i \in \{1, \dots, k+1\}$ and $e_{p_i} = (p_i, p_{i+1}) \in E$ for all $i \in \{1, \dots, k\}$. For road networks, a path is uniquely determined by its edge sequence and the notation can be simplified to

$$P = (e_{p_1 p_2} \dots e_{p_k p_{k+1}}).$$

A path set in G is defined as a set of path whose origin and length are identical in a road network. To specify the origin and length of the path, a path set is denoted by $P^K(e_0)$, where e_0 and K are the origin and length of the path, respectively. e.g., $P \in P^k(e_{p_1})$

Definition 4 (Congestion): Given an edge e , a time duration T , and a short term time period $d(\ll T)$, an edge e is said to be congested in T with d , if (C1),(C2),(C3) are satisfied at any $t \in T$, where

$$(C1) : \quad \bar{s}(e, t)_5 \leq \text{avg}(s(e, T)) \quad (1)$$

$$(C2) : \quad \bar{s}(e, t)_5 \leq \bar{s}(e, t)_{30} \quad (2)$$

$$(C3) : \quad s(e, t) \leq s(e, t-1) - \text{sd}(s(e, [t-d, t])). \quad (3)$$

The first condition in Eq.(1) indicates that $\bar{s}(e_0, t)_5$ should be smaller than the average speed for the given time duration T . The second condition in Eq.(2) is closely related to the concept of a death cross in stock chart analysis [7, 8], which is used to capture short-term speed patterns. It is known that a death cross appears on a chart when a moving average of short-term stock price crosses below its long-term moving average, which indicates the trend of recent price drop is greater than that of long-term. Likewise, we set the second condition whether the short-term (5 minute interval) speed trend overtakes the long-term (30 minute interval) speed trend. The last condition in Eq.(3) sets a condition whether there exists a radical speed drop. If a traffic incident occurs, the speed of vehicle affected by the incident is expected to drop sharply after the traffic incident. To reflect the variation of the speed distribution, the speed decreases by more than its standard deviation, $\text{sd}(s(e, [t-d, t]))$, is regarded as the sharp speed drop.

Definition 5 (Congestion indicator function): Given an edge e , a time duration T , and a short term time period $d(\ll T)$, a congestion indicator function is defined as

$$c_d(e, T) = \begin{cases} 1, & \text{if an edge } e \text{ is congested in } T \text{ with } d \\ 0, & \text{otherwise.} \end{cases}$$

The definition of congestion is defined in Definition 4. For simplicity, $c_d(e, T)$ can be abbreviated as $c(e, T)$, assuming $d = 10$.

Definition 6 (Congestion path): Given a path $P = (e_{p_1} e_{p_2} \cdots e_{p_k})$ in G and a time duration T , a congestion path is defined as finite sequence

$$c\text{-path}(P, T) = (c(e_{p_1}, T), c(e_{p_2}, T), \cdots, c(e_{p_k}, T)),$$

where $c(e, T)$ is congestion indicator function defined in Definition 5.

Definition 7 (Propagation path): Given a path $P = (e_{p_1} e_{p_2} \cdots e_{p_k})$ in G and a time duration T , a propagation path is defined as finite sequence

$$p\text{-path}(P, T) = (q(e_{p_1}, T), q(e_{p_2}, T), \cdots, q(e_{p_k}, T)),$$

where, for $i > 1$,

$$q(e_{p_i}, T) = \begin{cases} 1, & \text{if } c(e_{p_i}, T) = 1 \text{ and } c(e_{p_{i-1}}, T) = 1 \\ 0, & \text{otherwise.} \end{cases}$$

For $i = 1$, $q(e_{p_1}, T) = 1$ if $c(e_{p_1}, T) = 1$ and $q(e_{p_1}, T) = 0$ otherwise.

Definition 8 (Reported time): After a non-recurrent incident occurs, the point at which information about the incident is released to the public is defined as t_{rpt} . In the case of a traffic incident, it is usually difficult to specify when the incident occurred at point t_{rpt} , but it is certain that t_{rpt} is after that. It is known that it takes about 15 minutes for information to be released after an traffic incident occurs in South Korea.

IV Prioritizing Influential Factors for Traffic Incidents Impacts in a Large Scale Road Network

4.1 Introduction

There have been attempts to measure traffic incidents impacts and build incident impact prediction models for identifying influential factors. However, there are several limitations and difficulties in existing approaches.

First, most of existing approaches rely on quantitative measurements that are not measurable for non-recurrent traffic congestion in practice. For example, the incident duration, which is one of most popular measurements for incident impact [9], assumes that information about an incident time period is given [10]. Without such information, it is hard to define reasonable incident duration and to justify whole process of predicting and analyzing incident impact. The incident duration is measured by the Highway Capacity Manual as four phases [9]: incident detection time; incident response time; incident clearance time; incident recovery time. Among these four management phases, clearance time takes the longest duration [10]. Therefore, if such information is absent, it is hardly to define reasonable incident duration and to justify whole process of predicting and analyzing incident impact. In the case of South Korea, not only incident information collected by National Police Agency include no aforementioned information but is estimated largely based on the arbitrary decision relied upon one's experience rather than on statistical approaches. Therefore, devising a way of measuring it must be preceded to quantify the impact of an incident duration.

Second, existing studies have focused on temporal impact of incident impact, not providing any spatial or speed dynamics aspect of incident impact. When an incident occurs, traffic congestion inevitably follows, which affects near roads and causes additional congestion, especially to all the traffic leading to the congested roads. Such spatial impact of an congestion being spread is called congestion propagation. Depending on the severity of an incident, its impact being spatially propagated to multiple different roads varies. In addition, its risk may varies significantly across similar, or even identical, road geometric features. A spatial analysis of traffic incidents allows for the consideration of various spatial factors affecting congestion propagation. Furthermore, this prior knowledge can be usefully utilized in developing effective traffic planning or engineering solutions to reduce traffic crashes. For speed dynamics aspect of non-recurrent traffic congestion, the rate of change in the average speed of the road in the event of an incident can also be an indicator of the seriousness of the incident at that time. The more unexpected and serious the incident is and the more difficult it is to avoid, which lead the drivers on the road tend to have sharp drop in speed. Thus, predicting such spatial or speed dynamics aspect of non-recurrent traffic congestion provides valuable information for alleviating traffic congestion being propagated to nearby roads in advance via real-time response, furthermore, it is required to measure their impacts with new quantitative measurements.

Third, predicting non-recurrent congestion measures is notoriously difficult, because the underlying physics and interconnected congestion propagation processes is complex nonlinear turbulent dynam-

ical [11]. This mainly results from that the perfect knowledge of nature is never known in practice. Consequently, model error is inevitable in the development of congestion propagation prediction models and identifying influential factors based on such unreliable prediction models is meaningless.

Moreover, identifying influential factors for the non-recurrent traffic congestion impact and prioritizing them are our focus in this study. Most of the previous work has attempted to interpret the model in global manner and have features importance of incident impact [12]. By contrast, this study identifies influential factors based on local interpretation because of the following three reasons. First of all, we can extract and compare feature importance in the nonlinear models. Simple linear models are easy to understand but nonlinear model works better in the high-dimensional cases. In other words, nonlinear model often do a better job in predicting new instances correctly, but the reasons why a certain prediction was made is hidden. Second, we focus on the individual incidents to understand the relationship between variables and congestion measurements. This is because global interpretation of the model may not be useful in understanding individual nonrecurring incident because the topological features and the type of an incident are independent and identical. Furthermore, local approach help us to understand the model when the model makes wrong prediction, such as false negative [13]. Therefore, we adopted LIME (Local Interpretable Model-agnostic Explanations), which locally interpret the model to understand the important features for each incident impact indices by individual incident.

Thus, we propose a new methodology for measuring the impact of non-recurrent traffic congestion propagation and for prioritizing influential factors in the development of non-recurrent traffic congestion impact prediction models. The main contributions of our work are as follows:

- We develop three new quantitative metrics for non-recurrent traffic congestion: the incident duration, the congestion propagation level, and the speed drop ratio. Compared to the previous literature, these metrics are applicable even in the absence of accurate time information about the incident and measure both the temporal and spatial impacts of incidents.
- We apply Local interpretable model-agnostic explanations (LIME) in the development of congestion propagation prediction models to extract meaningful information about influential factors on the proposed metrics.

The remainder of this paper is organized as follows. Section 4.2 reviews related works in the literature. Section 4.3 proposes a new methodology for measuring non-recurrent traffic congestion and prioritizing influential factors on such measurements. The related experimental results are analyzed in Section 4.5.

4.2 Related Works

Incident Delay Estimation

Several methods have been developed to estimate total delay of incoming roads of incidents. These methods are categorized into two categories: 1) analytical methods using deterministic queuing diagrams [14–16]; 2) shock wave theory [17–19]. A queuing model is constructed so that queue lengths and

waiting time can be measured. Queuing model is one of the most widely used approaches to measure the delay caused by an incident. The theory of traffic shock wave determines delays time by measuring difference state of traffic average flow and density. [17] used shock wave theory to develop formulas for calculating individual and total delays of incoming roads during incident situation. The formulas are based on areas and densities of regions representing different traffic states. [18] compared queuing analysis and shock wave analysis for calculating total incident delay on a highway roads. It assumed a unique flow-density relationship and derived the equations of total delay. These methods have been either used independently or combined.

However, these approaches have limitations in measuring the impact of non-recurrent traffic congestion propagation. First, they rely on various traffic features such as traffic flow, density, and occupancy to quantify incident impact, which are not accessible in practice. Second, these approaches assume that the exact time of traffic accident occurrence and recovery is known, which is not true in reality.

Traffic Condition Indicators

A significant number of studies have measured urban traffic conditions in various ways using the multiple indicators that value traffic state. Speed Reduction Index (SRI) ratio was suggested by [20,21] to measure relative speed variation between congested state and free-flow condition; and defined congestion as the index value exceeds specific threshold. The Urban Congestion Report of 2019 published by FHWA (Federal Highway Administration) suggested the free-flow speed as the 85th percentile of the off-peak speed which refers the average speed during 9:00 a.m. to 4:00 p.m. and 7:00 p.m. to 10:00 p.m. from Monday through Friday, and Saturday and Sunday 6:00 a.m. to 10:00 p.m. [22]. [20] developed Speed Performance Index (SPI) using the maximum permissible speed as free flow.

Above methods define traffic congestion as their indicators rally under the certain threshold. However, these indicators overly simplified the actual incident duration period, the duration of the impact of a traffic incident. In addition, any incident that prominently affects neighboring roads but not exceed the threshold would be ignored if threshold speed limit determines start and recovery time of an incident. With these reasons, it is reasonable to capture incident oriented congestion by measuring relative speed decline compared to normal traffic state.

Feature Importance for Incidents Impact

Existing approaches modelling the incident impact can be generally divided into two mainstreams: statistical methods and machine learning methods. Most prevailing statistical approaches to model incident duration are regression models and hazard-based models. They are commonly based on the rigorous mathematical assumptions to have the capability of interpreting the relationship between estimator and explanatory variables [23]. Regression model includes linear/nonparametric regression [24–26]. Hazard-based models are superior in capturing duration effects [27–29] under the three main methods: proportional hazard model [30], accelerated failure time (AFT) model [28,31], and the Cox proportional hazards regression model [32,33]. However, since these statistical methods performs under the several

hypotheses and constraints, they often lose interpretation power and prediction power, which leads to the necessity of nonlinear based method, such as machine learning (ML).

ML approaches have been successfully employed to estimate and predict the traffic incident duration and interpret the model as they describe influential factors [12]. Especially, to identify features importance that contributes the incident impact, interpretability of the model is used to confirm other important desiderata of ML systems. There exist many auxiliary criteria that one may wish to optimize. Notions of fairness or unbiasedness imply that protected groups (explicit or implicit) are not somehow discriminated against [34]. Generally, feature importances can be divided into modular global and local importances [35,36]. While a modular global feature importance measures the importance of the feature for the entire model, a local importance measures the contribution of the feature for a specific observation. However, to reflect the uniqueness of an incident, we use the local interpretation methodology which is called LIME (Local Interpretable Model-agnostic Explanations) [37].

The LIME method interprets individual model predictions based on locally approximating the model around a given prediction. We now evaluate how faithful they are to the underlying model using LIME. One expectation is that our explanations should be locally accurate, i.e. in the vicinity of the input data point, our explanation should be faithful to the model [38,39].

4.3 Methodology

This section proposes three non-recurrent congestion measures to quantify the impact of traffic incidents, as well as a local interpretation method to prioritize incidents-related features that affect the impact of traffic incidents. Terminologies defined in Section III will be used.

Measuring Non-recurrent Traffic Congestion

To measure the road traffic congestion, numerous congestion measures have been developed considering different performance criteria. Depending on these criteria, the congestion measures can be categorized into six categories: (i) speed, (ii) travel time, (iii) delay, (iv) level of services (LoS), (v) congestion, and (vi) federal congestion indices [40]. In this study, we propose three new non-recurrent congestion measures: incident duration, speed drop ratio, and congestion propagation level. The main difference of the proposed measures from existing ones is that our measures focus on quantifying the congestion level driven by non-recurrent traffic events such as traffic incidents.

Fig. 2 illustrates the concepts of the proposed non-recurrent traffic congestion measures. x -axis and y -axis in the figure represent time index and normalized speed, respectively. Normalized speed means the value of speed divided by its maximum allowed speed for each road segment. Vertical red dashed lines represent the time when a traffic incident reported. Driven by the traffic incident, Fig. 2(a) introduces the concepts of incident duration and speed drop, while Fig. 2(b) shows the concept of congestion propagation to k -hop incoming neighbors. The black solid line indicates the speed pattern on the road segment e where a traffic incident occurred. Red, green, blue solid lines represent three road segments in the 1-hop incoming neighbors $N_{in}^1(e)$. That is, $|N_{in}^1(e)| = 3$ in this example. In addition,

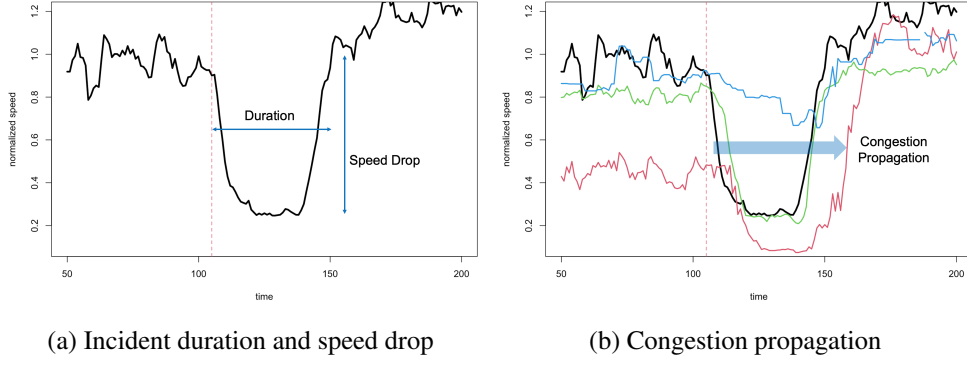


Figure 2: The concepts of the proposed non-recurrent traffic congestion measures

the speed data is recorded for every minute. The further hop incoming neighbors are not shown in this figure on purpose, but they will be included and analyzed in Section 4.5.

The three proposed measures are defined one by one as follows.

Incident Duration

Suppose that a traffic incident occurred on the road segment e_0 and reported at time t_{rpt} . To determine the incident duration, it is required to estimate the time when the incident starts and ends. For a traffic incident reported at t_{rpt} , we estimate the incident starting time, $t_{st}^{e_0}$ by

$$t_{st}^{e_0} = \min\{t \in [t_{rpt} - 120, t_{rpt} - 1] : c_{10}(e_0, [t - 120, t - 1]) = 1\}. \quad (4)$$

In this study, we set 120 minutes as a period to analyze the starting point of the incident in order to have enough time period that both include regular state and transition state. The previous time period is arbitrary but it has to be long enough to hold both aforementioned traffic states.

On the other hand, we estimate the incident recovery time, $t_{end}^{e_0}$, by

$$t_{end}^{e_0} = \min\{t \in [t_{rpt} + 1, t_{rpt} + 240] : s^{\text{REG}}(e_0) < \min(s(e_0, [t + 1, t + 10]))\}, \quad (5)$$

where

$$s^{\text{REG}}(e_0) = \text{avg}(s(e_0, [t_{st}^{e_0} - 120, t_{st}^{e_0} - 1])).$$

$s^{\text{REG}}(e_0)$ represents the average speed in the road segment e_0 before the traffic incident occurred. Eq.(5) means that $t_{end}^{e_0}$ is determined by the time when its speed is greater than $s^{\text{REG}}(e_0)$ 10 consecutive minutes. By Eq.(4) and Eq.(5), the incident duration ($Y_1^{e_0}$) can be found by

$$Y_1^{e_0} := t_{end}^{e_0} - t_{st}^{e_0}. \quad (6)$$

The time period, $[t_{st}^{e_0}, t_{end}^{e_0}]$, is denoted by $T_{Y_1}^{e_0}$. Note that the length of $T_{Y_1}^{e_0}$ is $Y_1^{e_0} (> 0)$.

Congestion Propagation Level

Given an incident duration, $T_{Y_1}^{e_0}$, defined by Eq.(6), and propagation paths $\{p\text{-path}(P_w, T_{Y_1}^{e_0})\}$, where

$$P_w = (e_{p_1}^w e_{p_2}^w \cdots e_{p_K}^w) \in P^K(e_0), w = 1, \dots, W$$

the *congestion propagation level* is defined by

$$Y_2^{e_0} := \frac{1}{W \cdot K} \sum_{w=1}^W \left(\sum_{i=2}^K q(e_{p_i}^w, T_{Y_1}^{e_0}) \right), \quad (7)$$

where W is the number of paths in $P^K(e_0)$, $W = |P^K(e_0)|$. According to Eq.(7), we exclude $q(e_{p_1}^w, T_{Y_1}^{e_0})$ for all w in calculating $Y_2^{e_0}$, because

$$q(e_{p_1}^1, T_{Y_1}^{e_0}) = \cdots = q(e_{p_1}^W, T_{Y_1}^{e_0}) = q(e_0, T_{Y_1}^{e_0}) = 1$$

by Propagation 1.

Lemma 1 Suppose that a non-recurrent traffic accident occurred on road segment e_0 and reported at t_{rpt} . Given propagation paths $P_w \in P^K(e_0)$, $w = 1, \dots, W$, and the incident duration time period $T_{Y_1}^{e_0}$,

$$q(e_{p_1}^1, T_{Y_1}^{e_0}) = \cdots = q(e_{p_1}^W, T_{Y_1}^{e_0}) = q(e_0, T_{Y_1}^{e_0}) = 1.$$

Proof 1 A path set is defined in Definition 3 as a set of path whose origin and length are identical in a road network. This implies

$$e_{p_1}^1 = e_{p_1}^2 = \cdots = e_{p_1}^W = e_0.$$

Since $t_{rpt} \in T_{Y_1}^{e_0}$, then $c(e_0, T_{Y_1}^{e_0}) = 1$ by Definition 5. Consequently, provided by Definition 6 that $q(e_0, T_{Y_1}^{e_0}) = 1$, we can conclude that

$$q(e_{p_1}^1, T_{Y_1}^{e_0}) = \cdots = q(e_{p_1}^W, T_{Y_1}^{e_0}) = q(e_0, T_{Y_1}^{e_0}) = 1.$$

Note that we only consider the K -hop incoming neighbors that are congested during the congestion duration, $T_{Y_1}^{e_0}$, to make it sure that the congestion in the incoming neighbors results from the traffic incident on the road segment e_0 .

The range of $Y_2^{e_0}$ becomes $0 \leq Y_2^{e_0} \leq 1$. $Y_2^{e_0} = 1$ means a traffic incident occurred on the road segment e_0 causes congestion on all road segments within $N_{in}^K(e_0)$, while $Y_2^{e_0} = 0$ means no congestion propagation at all.

Speed Drop Ratio

To quantify the impact of a traffic incident on speed reduction, we propose a new measurement, called the *speed drop ratio*. The speed drop ratio, $Y_3^{e_0}$, is defined by

$$Y_3^{e_0} := \max \left(0, 1 - \frac{\min(s(e_0, [t_{rpt} + 1, t_{rpt} + 60]))}{\max(s(e_0, [t_{rpt} - 60, t_{rpt} - 1]))} \right). \quad (8)$$

Y_3^{e0} , $0 \leq Y_3^{e0} \leq 1$, is the speed difference ratio of the maximum speed for 1 hour before t_{rpt} and the minimum speed for 1 hour after t_{rpt} . This is based on the assumption that the average speed drops sharply between 1 hour before and 1 hour after the incident report time.

A similar measurement in the literature is the speed reduction index (SRI) [40]. SRI is the ratio of the relative speed change between congested and free-flow conditions. SRI can be used to detect whether congestion occurs or not, while Y_3^{e0} can be used to identify non-recurrent speed dynamics.

Model Based Feature Importance

Local interpretable model-agnostic explanations (LIME) [37] is a novel explanation technique that explains the predictions of any prediction model by learning a local surrogate model. Local surrogate models are interpretable models that are used to explain individual predictions of any prediction model. Suppose that f_k is a prediction model to predict Y_k , $k = 1, 2, 3$, $Y_k = f_k(X)$, where X is a feature matrix. Here, it is assumed that we can probe the prediction model f as often as we want. Let a model $g \in G$ is an explanation model, where G is a class of potentially interpretable models. Our goal is to understand why the prediction model makes a certain prediction through a local surrogate model. LIME produces the local surrogate model by

$$\xi_k(x) = \arg \min_{g \in G} \mathcal{L}(f_k, g, \pi_x) + \Omega(g), \quad (9)$$

where $\mathcal{L}$ be a measure of how unfaithful g is in approximating f in the locality defined by π_x and $\Omega(g)$ be a measure of complexity of the explanation $g \in G$. Here, we use the locally weighted square loss $\mathcal{L}$, the exponential kernel π_x , and the complexity measure $\Omega(g)$ as defined in [37]. In order to learn the local behavior of f , $\mathcal{L}$ is approximated by drawing samples, weighted by π_x . Samples are perturbed instances around x_i by adding random noise to the local instances near x_i . Model g is trained with perturbed instance z_i , which is generated by adding random noise to the local instances near x_i . Coefficient is trained by ridge regulation by adding $\lambda \sum_{j=1}^P W_j^2$ to squared loss [41],

$$\text{minimize} \left(\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^P W_j^2 \right),$$

which is controlled by tuning parameter λ and P is the number of features. The ridge regression model pushes many of the correlated features toward each other rather than allowing for one to be wildly positive and the other wildly negative. This help to clarify the importance of features by comparing I_j .

In this paper, for example, ridge regression models are chosen as interpretable surrogate models G . When using ridge regression models as explanations, for an instance x_i and explanation $g_i = \xi_k(x_i)$, we set $H_{ij} = |h_{g_i,j}|$, where $h_{g_i,j}$ is the coefficient corresponding to j th feature in g_i . For the j th feature in H , its global importance I_j can be obtained by

$$I_j = \frac{\sum_{i=1}^n H_{ij}}{n} \quad (10)$$

4.4 Case Study: Incident on Gangbyeonbuk-ro

The proposed non-recurrent traffic congestion measures are applied with a real incident case on Gangbyeonbuk-ro in Seoul. A traffic incident was reported at 22:16 on Sep 4, 2020 as shown in Fig. 3. This road network is known for high traffic and high complexity. The asterisk in the figure indicates the exact location where the incident occurred and the black colored line (A) indicates the road segment where the location belongs to. Similarly, red colored lines indicate 1-hop incoming neighbors (B, H, E) of the incident road (A), and blue and green colored lines represent 2-hop incoming neighbors (C, I, K, F) and 3-hop incoming neighbors (D, J, L, M, G), respectively.



Figure 3: A road network with a real incident in Seoul

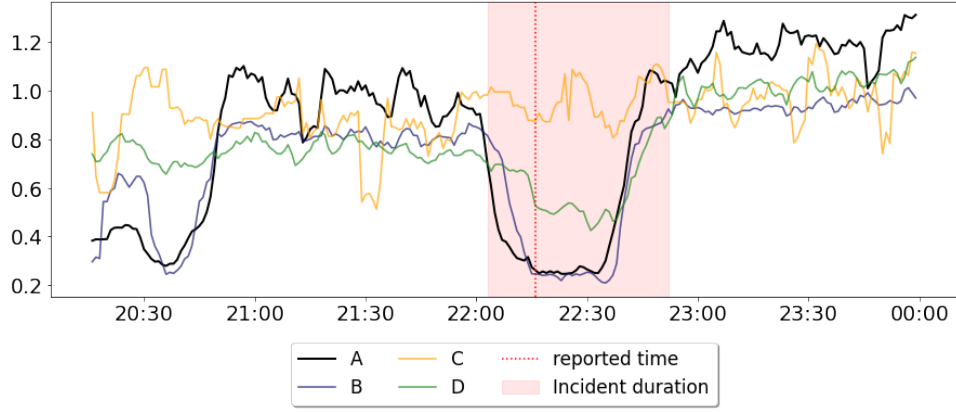
Consequently, the influence of the traffic from the road A stretches to five different incoming paths within three hops. The corresponding path set is

$$P^3(A) = \{(A, B, C, D), (A, H, I, J), (A, H, K, L), (A, H, K, M), (A, E, F, G)\}.$$

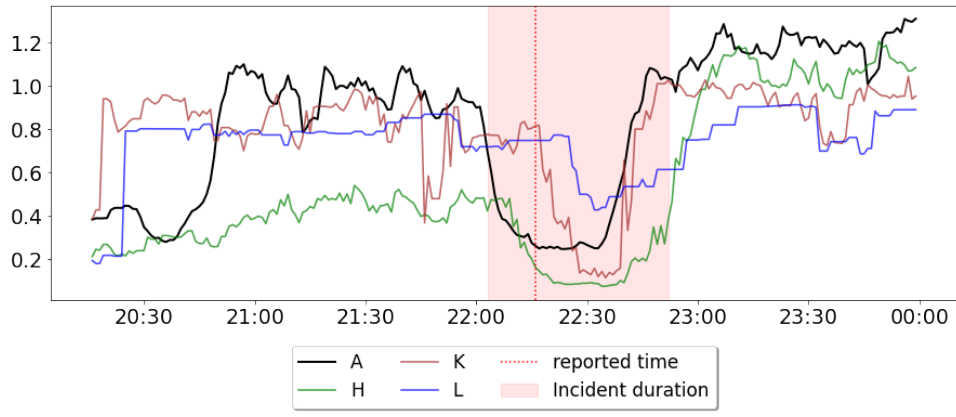
Fig. 4(a) and Fig. 4(b) illustrate speed patterns of the first path (A, B, C, D) and the third path (A, H, K, L) in the path set $P^3(A)$. y-axis represents the speed values normalized by their maximum speed limit, 80 km/h. Fig. 4(a) illustrates that only the road B had been impacted from the incident, while Fig. 4(b) shows that the impact of the incident propagates sequentially as hop increases. As the first path, (A, B, C, D) leads to the Stadium. Second, third, and fourth road, (A, H, I, J), (A, H, K, L), (A, H, K, M) shares the 1st incoming hop road H. However, while second path leads to inner ring road, third and fourth paths lead to inner city. Last path, (A, E, F, G), remains in Gangbyeonbuk-ro.

For the path set, the proposed method was applied to measure the non-recurrent traffic congestion. First, as proposed in Eq.(6), the incident duration Y_1^A is computed to be 49 minutes, where $t_{st}^A = 22:53$ and $t_{end}^A = 22:04$. The estimated incident duration is marked as red box in Fig. 4.

Second, we calculated the propagation path for the path set $P^3(A)$ based on the time period of the



(a) Speed curves of 1st path



(b) Speed curves of 3rd path

Figure 4: Speed curve for each path

incident duration, $T_{Y_1}^A = [t_{st}^A, t_{end}^A]$ as follows.

$$\begin{bmatrix} p\text{-path}((A, B, C, D), T_{Y_1}^A) \\ p\text{-path}((A, H, I, J), T_{Y_1}^A) \\ p\text{-path}((A, H, K, L), T_{Y_1}^A) \\ p\text{-path}((A, H, K, M), T_{Y_1}^A) \\ p\text{-path}((A, E, F, G), T_{Y_1}^A) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

The propagation path yields $Y_2^A = 0.87$ as defined in Eq.(7). Note that all incoming neighboring road segments except for road C were experiencing relatively significant speed drop within calculated incident duration. It is worthy to note that the road segment D turned out to be congested ($c(D, T_{Y_1}^A) = 1$), but the result of $p\text{-path}((A, B, C, D), T_{Y_1}^A)$ indicates the congestion at D was not propagated from its preceding road C .

Last, the speed drop ratio Y_3^A described in Eq.(8) was computed. The blue and red lines indicate the maximum speed on the road A for 1 hour before the reported time and the minimum speed for 1 hour after the reported time, respectively. Based on $\max(s(A, [t_{st}^A - 60, t_{st}^A])) = 1.092$ and $\min(s(A, T_{Y_1}^A)) = 0.246$, Y_3^A is computed to be 0.77

Consequently, the proposed non-recurrent congestion measures of the incident are computed to be

$$(Y_1^A, Y_2^A, Y_3^A) = (49, 0.87, 0.77).$$

4.5 Real Data Example: Traffic Incidents in Seoul

The proposed method was validated with 390 real traffic incidents occurred from September 2 to December 1, 2020, in Seoul, South Korea. Fig. 5 depicts the locations of filtered 390 traffic incidents (red dots) considered in this study.

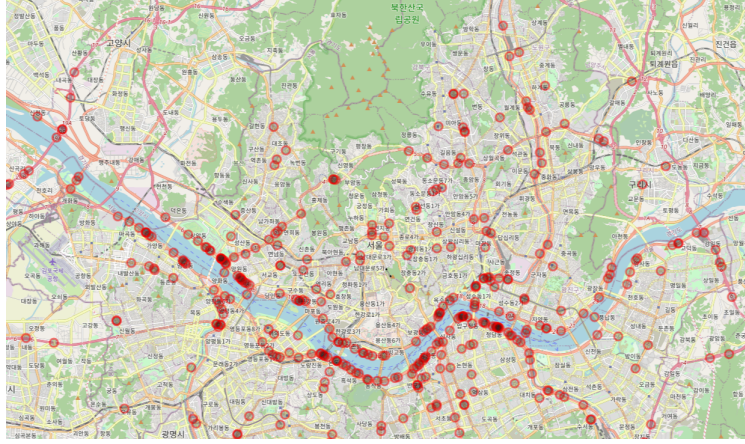


Figure 5: Incidents location in Seoul (from 02/09/2020 to 01/12/2020)

Datasets

For validation purpose, two different types of datasets were used from different sources: the traffic dataset provided by the corporation NAVER navigation team and the incident dataset provided from Korea National Police Agency.

The traffic dataset consists of traffic road network, trajectory based speed, weather, and road attributes of major metropolitan area, city of Seoul, where nearly half of the entire country's population reside. The speed data were described in GPS trajectory manner. A GPS trajectory consists of a series of points with latitude, longitude, and timestamp information generated when navigation users travel. However, this data does not represent the graph based speed variation over time, but each individual's travel trajectory for specific period of time. Therefore, we had to undergo map-matching process [42] to align a sequence of observed user positions with the road network and obtain average speed, $\{s(e, t) \in S : E \times T \rightarrow \mathbb{R}^+\}$, for each road segment, $e \in E$ at different time point $t \in T$.

The incident dataset used were provided from Korea National Police Agency containing 1,425 incidents occurred during the same time period in Seoul. Each incident record includes reported time, description of the incidents, originating location described in both geographical coordinates and road segment ID (called SID) which corresponds to $e \in E$. Among 1,425 incidents, we selected the 390 incidents whose dynamic speed drop is less than 30% for validation purpose.

Independent Variables

Table 1 summarizes the key independent variables for incident impact analysis. These variables are features extracted from the traffic dataset because they are relevant with incident impact prediction. Extracted features can be categorized into five categories: temporal, topological, speed, operational, and weather.

Table 1: Description of independent variables

Categories	Variables	Value set & Description
Temporal	Time in day	1: $0 \leq \text{incident time} < 6$ 2: $6 \leq \text{incident time} < 12$ 3: $12 \leq \text{incident time} < 18$ 4: $18 \leq \text{incident time} < 24$
	Day in week	0: Weekday 1: Weekends (Saturday and Sunday)
Topological	Length	Length of the road segment where incident occurred
	Number of lane	Number of lanes of an incident road
	Number of incoming road	$ \sum_{k=1}^K N_{in}^k(e_0) $
	Number of outgoing road	$ \sum_{k=1}^K N_{out}^k(e_0) $
	In-Degree centrality	Cd_{in}
Speed	Out-Degree centrality	Cd_{out}
	Incoming correlation	Speed correlation between incident road and incoming roads
	Outgoing correlation	Speed correlation between incident road and outgoing roads
	Incident road speed limit	Road maximum speed limit
Operational	Speed dynamics	Speed drop ratio before an incident reported
	Information source	Information source of incident reporting
		1: Police Station
		2: Traffic broadcasting
		3: National Transport Information Center
Weather	Rain	4: Infrastructure and Transport
		0: No rain 1: Rain

1. *Temporal*: Two variables are considered: ‘Time in day’ and ‘Day in week’. The former represents the daily pattern and the latter is related to the weekly patterns.
2. *Topological*: The topological features of the road network include ‘Length’, ‘Number of lane’, ‘Number of incoming road’, ‘Number of outgoing road’, ‘In-Degree centrality’, and ‘Out-Degree centrality’. ‘Length’ is the geographical length of road segment where incident occurred. ‘Number of lane’ is the number of lanes present on an incident road. ‘Number of incoming road’ and ‘Number of outgoing road’ equal to $|\sum_{k=1}^K N_{in}^k(e_0)|$ and $|\sum_{k=1}^K N_{out}^k(e_0)|$, respectively. Lastly, the degree centrality measures the complexity of e_0 . Originally, the concept of the degree centrality

was introduced to measure the complexity of a target node in undirected network [43–45]. In this study, we modified the original concept to measure the complexity of a target edge, e_0 , in a directed network. In a directed network, we measured the in-degree centrality (Cd_{in}) and the out-degree centrality (Cd_{out}), separately, as follows:

$$Cd_{in} = \frac{|N_{in}^1(e_0)|}{|\sum_{k=1}^K N_{in}^k(e_0)| + |\sum_{k=1}^K N_{out}^k(e_0)|} \quad (11)$$

$$Cd_{out} = \frac{|N_{out}^1(e_0)|}{|\sum_{k=1}^K N_{in}^k(e_0)| + |\sum_{k=1}^K N_{out}^k(e_0)|}. \quad (12)$$

3. *Speed*: Incoming and outgoing correlation represent the Pearson correlation between the incident road and neighboring roads. When an incident occurs, the traffic flow between the incident road and neighboring roads can be positively or negatively correlated. To measure such correlations, we define incoming correlation ($corr_{in}(e_0, K)$) and outgoing correlation ($corr_{out}(e_0, K)$) by

$$corr_{in}(e_0, K) = \frac{\sum_{k=1}^K \sum_{e_{in} \in N_{in}^k(e_0)} \rho(e_0, e_{in} : T_{Y_1}^{e_0})}{\sum_{k=1}^K |N_{in}^k(e_0)|}$$

$$corr_{out}(e_0, K) = \frac{\sum_{k=1}^K \sum_{e_{out} \in N_{out}^k(e_0)} \rho(e_0, e_{out} : T_{Y_1}^{e_0})}{\sum_{k=1}^K |N_{out}^k(e_0)|},$$

where $\rho(e_A, e_B; T)$ is the correlation coefficient of two time series, $s(e_A, T)$ and $s(e_B, T)$. Also, we used the maximum speed of the incident road as an independent variable. Speed dynamics (Sd) is another speed-related independent variable that measures the degree of a speed drop over an hour prior to an incident,

$$Sd = 1 - \frac{\min(s(e_0, [t_{rpt} - 60, t_{rpt} - 1]))}{\max(s(e_0, [t_{rpt} - 60, t_{rpt} - 1]))}. \quad (13)$$

4. *Operational*: Operational variables include the source of incident report. Depending on the party detecting an incident, the time for cleaning up the incident scene might varies [46]. In our data, there are four main sources: Police Station, Traffic broadcasting, National Transport Information Center, Infrastructure and Transport.
5. *Weather*: When an incident occurs, weather may impact on the duration and recovery of the incident. Due to the limitations of data collection, this study only considered whether it was raining.

4.6 Prioritizing Influential Factors for Traffic Incidents Impacts

Suppose that f_k is a prediction model to predict Y_k , $k = 1, 2, 3$, $Y_k = f_k(X)$, where X is a feature matrix comprising of independent variables in Table 1 for 390 traffic incidents. Independent variables were normalized and used as machine learning model features. Multiple machine learning models were applied the prediction model f including Linear Regression (LR), Lasso Regression (Lasso), Ridge Regression (Ridge), Support Vector Machine with RBF kernel (SVR), Decision Tree (DT), Random

Forest (RF), XGBoost, LGBM, MLP, and Ensemble model. For each Y_k , we constructed an ensemble model with the three best performing models.

The datasets are randomly divided into a training set (300 incidents) and a test set (90 incidents). For the training set, 5-fold cross validation was used to select the best model base on the mean absolute percent error (MAPE) and mean absolute error (MAE). Details of the all regression model performance are shown in Table 2.

Table 2: Model comparison

	Incident duration		Propagation level		Speed drop ratio	
Name	MAPE	MAE	MAPE	MAE	MAPE	MAE
LR	0.675	57.923	0.334	25.244	0.100	0.069
Lasso	0.627	52.621	0.350	25.419	0.128	0.091
Ridge	0.656	55.706	0.351	25.791	0.098	0.068
SVR	0.506	52.679	0.359	24.677	0.112	0.077
DT	0.909	78.344	0.532	35.183	0.131	0.091
RF	0.692	58.706	0.367	26.545	0.098	0.069
XGBoost	0.750	65.558	0.399	29.247	0.106	0.073
LGBM	0.680	61.226	0.363	26.029	0.105	0.072
MLP	0.623	52.372	0.366	27.003	0.124	0.084
Ensemble	0.593	53.501	0.355	25.987	0.097	0.068

We used a ridge regression model $g \in G$ as an explanation model, where G is a class of potentially interpretable models. Our goal is to understand why the prediction model makes a certain prediction through a local surrogate model using Eq.(9).

Table 3: Feature importance for target variables

Categories	Variables	Incident Duration		Propagation Level		Speed Drop Ratio	
		Mean	Relative(%)	Mean	Relative(%)	Mean	Relative(%)
Temporal	Time in day	0.210	0.63	0.444	2.52	0.002	2.37
	Day in the week	0.034	0.10	0.089	0.51	0.000	0.00
Topological	Length	2.267	6.84	2.171	12.31	0.003	3.78
	Number of lanes	0.455	1.37	0.105	0.60	0.002	2.38
	Number of incoming roads	5.493	16.57	2.718	15.06	0.002	2.64
	Number of outgoing roads	7.501	22.62	1.222	6.93	0.000	0.36
	In-Degree centrality	1.147	3.45	2.718	15.41	0.009	9.91
	Out-Degree centrality	0.805	2.43	0.084	0.48	0.007	6.99
Speed	Incoming Correlation	5.964	17.99	1.452	8.23	0.002	1.84
	Outgoing Correlation	0.926	2.79	2.265	12.84	0.002	2.23
	Incident road speed limit	1.987	5.99	2.099	11.90	0.006	6.48
	Speed dynamics	4.360	13.15	0.881	5.00	0.058	58.37
Operational	Information source	0.176	0.53	0.196	1.11	0.00	0.42
Weather	Rain	1.830	5.52	1.254	7.11	0.002	2.24

Using the local surrogate model produced by LIME, we can identify the important features for

individual and obtain the global importance by Eq. (10). Table 3 show the global importance of each feature for incident duration, propagation level, and speed drop ratio. In the table, the relative feature importance is also calculated by dividing the sum of all feature importance.

For incident duration prediction, ‘Number of outgoing road’ contributes dominantly with relative importance of 22.62%. Outgoing road is a road that is going away from the target road, incident road in this case. In other word, the more outgoing roads are connected to the incident road, the more possible gates for the drivers to escape from the scene, which eventually shorten the duration of incident impact. Therefore, as a topological feature, the ‘Number of outgoing roads’ plays significant role for forecasting incident duration.

For propagation level, the ‘In-Degree centrality’ is the most contributing variable with the relative importance of 15.41%. As described in Eq. (11), this variable represents how densely first hop incoming roads are heading toward the incident road. The higher its ‘In-Degree centrality’ value is, the more possible incoming road segments that are potential to receive congestion impact from the incident road. Therefore, this implies that depending on the ‘In-Degree centrality’ value, the number of candidate paths that could be influenced by the congestion generated from the incident road may vary. Likewise, the ‘Number of incoming roads’, following most relative variable with relative importance of 15.06%, can provides similar insight for predicting propagation level.

For speed drop ratio, ‘Speed dynamics’ is the most dominant factor with 47.69% relative importance. This result is very obvious because the time when the incident was reported is a situation in which the incident has already occurred and has progressed to some extent. Even after the incident is reported, there may be a decrease in speed, but the degree is likely to be insufficient. Therefore, the speed dynamic is sufficient enough to represent total speed drop ratio during the incident duration.

We also observed that temporal, operational, and weather related features have comparably irrelevant impact on incident impact measurements. On the other hand, interestingly, topological variables contributes most in predicting incident duration, propagation level with higher relative importance. To best our knowledge, this finding has not been yet discovered from previous studies.

V A Propagation Prediction Method for Non-recurrent Traffic Congestion

5.1 Introduction

Predicting non-recurrent propagation patterns in an online manner is especially important for navigation service providers because they must provide new routes for drivers who are in or heading toward an incident area to help them avoid congested roads. However, it is challenging to model non-recurrent congestion propagation using typical machine learning methods due to the non-stationarity of speed patterns during incidents and lack of training incident data for a given area. These characteristics do not align with the typical assumptions adopted in time series modeling, such as the stationarity or abundance of a dataset [47]. Therefore, although numerous attempts have been made to predict recurrent propagation patterns (e.g., rush hour traffic) [48, 49], there have been few attempts to predict propagation patterns after non-recurrent events such as traffic incidents.

In contrast, "pattern matching" does not require any assumptions and is based on the imitation of historical patterns. Pattern matching methods identify historical patterns and assume that the patterns in current circumstances will evolve in a similar manner to the identified patterns. In this study, by utilizing the concept of pattern matching, we developed a novel method that predicts non-recurrent congestion propagation patterns by identifying recurrent congestion propagation patterns that have occurred in the past. The underlying assumption of the proposed method is that current non-recurrent propagation patterns will be similar to historical recurrent congestion patterns if the speed patterns of an incident road in the present and past are similar in scenarios where traffic network topology conditions are constant. We use dynamic time warping (DTW) to match the patterns of speed time series between the past and present by measuring the similarity between two time series.

Additionally, the proposed method enables incident-based online learning. Online learning refers to the process of updating a model as new data such as different traffic network regions or new speed patterns arrive over time. Traditional machine learning and deep learning methods fix specific traffic networks and train models only for those regions, which limits the learning of new locations and time periods [50]. Considering that incidents occur spontaneously in multiple areas and must be monitored over time, such methods are not suitable for real-time scenarios. However, our model can be applied in real-world scenarios anywhere and anytime because it performs point-in-time pattern matching for individual incidents.

The main contributions of our work can be summarized as follows:

- We propose a novel online learning method based on pattern matching to predict the propagation patterns of non-recurrent traffic congestion.
- We hypothesize and demonstrate that historical recurrent congestion pattern information is useful for predicting real-time non-recurrent congestion patterns.
- We validate the proposed method using 215 real traffic incidents and related user trajectory data

obtained from a major GPS navigation system in South Korea.

The remainder of this paper is organized as follows. Section 5.2 reviews related works on general approaches to recurrent congestion propagation prediction and pattern matching used for congestion analysis. Section 5.3 proposes a novel methodology for predicting the propagation patterns of non-recurrent traffic congestion. Section 5.4 presents related experimental results that justify the application of the proposed model to a real-world navigation system.

5.2 Related Works

Congestion Propagation

Related studies on propagation patterns have mainly focused on identifying frequently recurring congestion propagation footprints in the past. [51] proposed the STOTree algorithm and a frequent subtree algorithm for finding a frequent propagation tree of abnormal traffic flow between regions. [52] extended this work to develop a visual interface to explore traffic congestion and propagation patterns. [48] predicted the footprints of recurrent congestion propagation patterns in the form of propagation graphs called Pro-Graphs. [53] compared various traffic congestion prediction methods using processed GPS trajectory data and concluded that deep learning models obtain higher accuracy compared with conventional machine learning models. However, none of these studies have attempted to predict non-recurrent congestion propagation.

The other type of research related to our work is the prediction of near-future traffic conditions based on non-recurring events. [54] utilized the cell transmission model and proposed a simulation-based non-recurrent congestion propagation model, which was incorporated with information gained from advanced traveler information systems (ATIS). This demonstrated that information provided by ATIS is helpful for drivers, particularly during non-recurrent congestion caused by severe incidents. [55] proposed two models to predict next-time-step traffic volumes on a single road segment to characterize congestion based on current inflow, outflow, and traffic volumes on the target road segment and adjacent upstream road segments. [56] predicted propagation speeds and boundaries based on a shockwave model for congestion following an observed traffic incident.

Traffic Speed Pattern Recognition

The pattern matching and pattern imitation methods presented in [47, 57] are based on local modeling. Such methods avoid common assumptions that are required in the context of time series modeling, such as stationarity or Gaussian conditions, and are based on the imitation of patterns in historical data. They attempt to identify predefined patterns in past data histories neighboring the current time and assumes that the current scenario will evolve in a manner similar to the historical scenario. This method has several advantages in that it does not require prior knowledge or parameter identification, making it suitable for nonlinear and uncertain systems. Additionally, it has been empirically observed that results improve with sufficient historical data [58].

Pattern matching methods with DTW have been widely used in the field of traffic for clustering congestion by comparing pattern differences between normal traffic and congested traffic. [59] estimated the traffic on neighboring roads around a congested road by comparing barometer sensor readings with a pre-stored road signature utilizing DTW. [60] proposed a trajectory pattern matching method based on DTW and spectral clustering to detect abnormal traffic flows. [61] utilized DTW distance to analyze the similarity between vehicle trajectories and clustered vehicle trajectories based on the similarity scores. However, DTW has never been applied to match patterns between non-recurrent congestion from the present and recurrent congestion from the past. In this paper, we propose a method for predicting future non-recurrent congestion propagation patterns by discovering similar patterns in historical time periods.

Real-time Congestion Prediction

The essential purpose of traffic management systems is to provide drivers with timely and reliable information regarding incidents [62]. Therefore, traffic management systems should utilize models that operate in real-time scenarios.

Traffic data are collected from streaming data sources and used for traffic modeling [63,64]. Endless vehicular data are produced by inductive loops, GPS units in vehicles, traffic cameras, and infrared sensors. Therefore, real-time machine learning approaches should be considered in the traffic domain. Real-time machine learning (also known as stream learning or stream data mining) has acquired special relevance with the arrival of the big data era [65, 66]. Numerous attempts have been made to utilize real-time stream data. [67, 68] used real-time GPS data equipped in a floating car to detect the traffic conditions of road network. [69] implemented a GPS based road traffic congestion reporting system using Smartphone, equipped with a GPS receiver to probe congestion on the roads. [70] presented an expert system for detecting traffic congestion and incidents from real-time GPS data collected from GPS trackers or drivers' smartphones.

Compared to existing methods, we focus on the problem of congestion propagation prediction. To the best of our knowledge, this is the first attempt to develop a method of predicting congestion propagation based on GPS data in real-time scenarios.

5.3 Methodology

We propose a novel propagation prediction method for non-recurrent traffic congestion. Fig. 6 presents an overview of the proposed method.

Problem Statement

We consider a road e_0 that is reported to have had an incident at time t_r . The problem of interest in this study is the prediction of the congestion propagation patterns through the K -hop incoming neighbors of e_0 . Here, to predict congestion propagation patterns, we determine the propagation paths for all paths coming into e_0 , as defined in Definition 7.

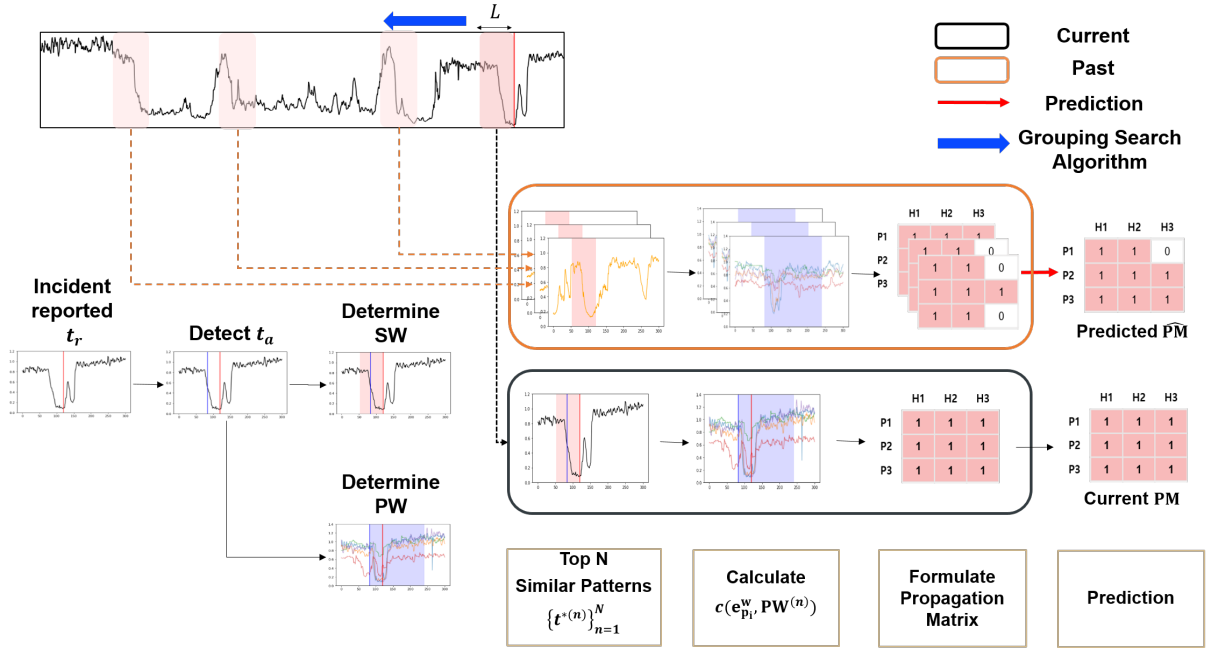


Figure 6: Methodology overview

Formally, given a time interval T during which congestion propagates and $\forall P_w = (e_0 e_{p_1}^w e_{p_2}^w \cdots e_{p_K}^w) \in P^K(e_0)$, we predict $p\text{-path}(P_w, T)$ as

$$\widehat{p\text{-path}}(P_w, T) = (\hat{q}(e_{p_1}^w, T), \hat{q}(e_{p_2}^w, T), \cdots, \hat{q}(e_{p_K}^w, T)), \quad (14)$$

where $\hat{q}(e_{p_i}^w, T)$ is the predicted value of $q(e_{p_i}^w, T)$. We predict $q(e_{p_i}^w, T)$ because the actual time interval T contains a future time span, which makes our problem difficult.

According to Definition 7, $q(e_{p_i}^w, T)$ indicates whether $e_{p_i}^w$ is propagated in T and can be obtained as $c(e_{p_i}^w, T)$. Consequently, to predict $q(e_{p_i}^w, T)$, we must predict $c(e_{p_i}^w, T)$. Therefore, we propose a prediction method for predicting $c(e_{p_i}^w, T)$ using historical speed data on e_0 . Additionally, we discuss how to determine an appropriate time interval T , which is referred to as the *propagation window (PW)* in this paper.

Similar Pattern Searching

To predict the $c(e_{p_i}^w, T)$ caused by a traffic incident on e_0 , we must search for traffic patterns that are similar to the non-recurrent traffic patterns generated by the incident. To search for traffic patterns, it is important to determine a reasonable time window for capturing traffic patterns. The time window is referred to as the *searching window (SW)* in this paper,

$$SW = [t_r - L : t_r]. \quad (15)$$

Let $s(e_0, [t_r - L : t_r])$ denote a *target traffic pattern*. Given the historical speed data on e_0 , $s(e_0, [1 : t_r])$, we wish to find similar patterns with the same time interval. Let $s(e_0, [t - L : t])$ and $t = L + 1, L + 2, \dots, t_r - L$ represent the candidates of similar patterns in the historical speed data, which are referred

to as *source traffic patterns*. To define the similarity between a source traffic pattern and target traffic pattern, we adopt DTW [71], which has been widely used to measure the similarity between pairs of time series. Let $dtw(\cdot, \cdot)$ denote the DTW distance functions. Then, given a source traffic pattern and target traffic pattern, the DTW distance can be obtained as

$$dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t])).$$

Therefore, we can find $s(e_0, [t^* - L : t^*])$, which is the most similar source traffic pattern to the target traffic pattern $s(e_0, [t_r - L : t_r])$, by solving the following optimization problem:

$$t^* = \arg \min_{t=L+1}^{t_r-L} dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t])). \quad (16)$$

However, congestion propagation prediction based on the most similar source traffic pattern is inevitably limited in terms of accuracy and robustness. Therefore, we consider selecting the N most similar source traffic patterns. However, selecting the top- N source traffic patterns is difficult because many locally similar traffic patterns exist around the most similar patterns. For example, $s(e_0, [(t^* - L + 1) : (t^* + 1)])$ may have a near-optimal similarity distance compared to $s(e_0, [t^* - L : t^*])$. Such locally similar traffic patterns contain no additional information for congestion propagation prediction.

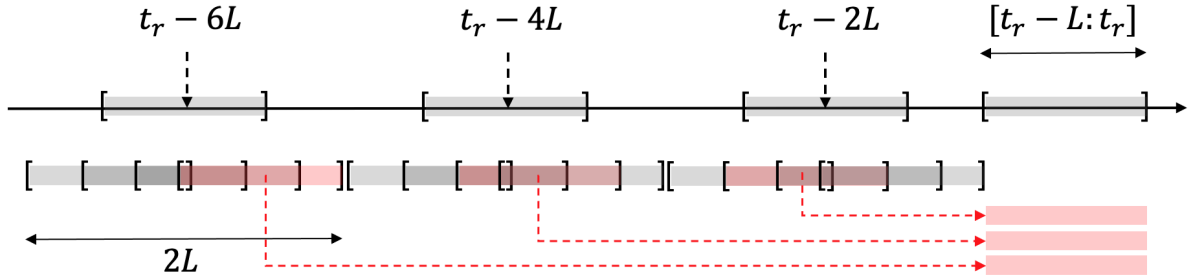


Figure 7: Grouping search algorithm

To exclude locally similar patterns from consideration effectively, we introduce a grouping search algorithm. The concept of the grouping search algorithm is illustrated in Fig. 7. One can see that the historical speed data on e_0 , $s(e_0, [1 : t_r])$ are split into multiple non-overlapping time periods

$$\begin{aligned} & \{[t_r - 2L : t_r - L], \dots, \\ & [t_r - (m+1)L : t_r - mL], \\ & \dots, [1 : 1 + L]\}. \end{aligned} \quad (17)$$

Suppose that there are M groups of non-overlapping time periods with lengths of $2L$ in Eq.(17). Then, for the m th non-overlapping time period, we apply Eq.(16) and obtain the most similar source traffic pattern within the time period as follows:

$$t^{*(m)} = \arg \min_{t=t_r-mL}^{t_r-(m+1)L} dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t])). \quad (18)$$

Based on Eq.(18), the grouping search algorithm yields a set of N time points

$$\{t^{*(n)}\}_{n=1}^N, \quad (19)$$

where $t^{*(n)}$ is the time point at which $s(e_0, [t^{*(n)} - L : t^{*(n)}])$ is the source traffic pattern that is the n th most similar to the target traffic pattern among $\{t^{*(m)}\}_{m=1}^M$. The grouping search algorithm is summarized in Algorithm 1.

Algorithm 1 Grouping search algorithm

Input: $t_r, L, N, s(e_0, [1, t_r])$

Output: S

```

1: for  $m = 1 : M$  do
2:    $D \leftarrow \emptyset, group.dist, t^*$  // Initialization
3:   for  $t = (t_r - (m + 1)L) : (t_r - mL)$  do
4:     if  $c(e_0, [t - L : t]) == 1$  then
5:        $dist \leftarrow dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t]))$ 
// calculate DTW distances
6:       if  $dist < group.dist$  then
7:          $group.dist \leftarrow dist$ 
8:          $t^* \leftarrow t$ 
9:       end if
10:    end if
11:  end for
12:   $D \leftarrow D \cup \{t^* : group.dist\}$ 
// dictionary item={key:value}
13: end for
14:  $D' \leftarrow sort(D)$  // sort  $D$  in ascending order by value
15: return  $D'.key[1 : N]$  // return keys of the first  $N$  items

```

There are three major traffic phases in congestion traffic flow: ‘free-flow phase’, ‘synchronized phase’, ‘congested phase’ [72]. The synchronized phase is a transitional traffic phase in which the flow of vehicles increases and the speed of them decreases before a congestion occurs. Particularly, for a non-recurrent congestion caused by a traffic incident, the synchronized phase indicates the occurrence of the traffic incident [73]. These phases reflect how the traffic pattern varies from the normal state to the state of being affected by the incident. Therefore, it can be stated that the ideal time window $[t_r - L : t_r]$ covers all three phases. Alternatively, L should be set to ensure coverage.

Because the appropriate length of L can be different for each traffic incident, we redefine Eq.(15) using the incident occurrence time. Let t_a denote the time point at which an incident occurs. Then,

$$SW = [t_r - L : t_r] = [t_a - PT : t_r], \quad (20)$$

where PT is a user-specific hyper-parameter that represents the length of the free-flow phase. A method of predicting t_a is proposed in the following section.

Propagation Window

To predict $c(e_{p_i}^w, T)$, we must rigorously determine the appropriate time window T . To this end, we define the propagation window PW as the time span over which the congestion caused by an incident e_0 is expected to propagate to the incoming roads $e_{p_i}^w \in \{N_{in}^k(e_0)\}_{k=1}^K$. Then,

$$PW = [t_a : t_r + F]$$

is the time span from t_a to $t_r + F$, where F is a user-defined hyper-parameter, future time range to know the pattern of congestion propagation after reporting an incident.

However, PW is difficult to compute because t_a is unknown or cannot be specified accurately at time t_r . It takes approximately 15 min on average for information to be released after a traffic incident occurs in South Korea. However, this period can vary significantly depending on various circumstances.

Therefore, we use the incident occurrence time detection method defined in 4

By using the detected incident occurrence time $\hat{t}_a$ in Eq.(4), PW can be rewritten as

$$PW = [\hat{t}_a : t_r + F] = [t_r - (t_r - \hat{t}_a) : t_r + F]. \quad (21)$$

The detection algorithm for the incident occurrence time is summarized in Algorithm 2.

Algorithm 2 Incident occurrence time detection

Input: $t_r, s(e_0, [t_r - 120 : t_r - 1])$

Output: t_a

```

1:  $t_a \leftarrow t_r$  // Initialization
2: for  $t = (t_r - 120) : (t_r - 1)$  do
3:   if  $t$  satisfies  $(C1), (C2), (C3)$  then
4:     if  $t < t_a$  then
5:        $t_a \leftarrow t$ 
6:     end if
7:   end if
8: end for
9: return  $t_a$ 

```

Propagation Prediction

For the N most similar source traffic patterns, we define $PW^{(n)}$ using $\{t^{*(n)}\}_{n=1}^N$ as follows:

$$PW^{(n)} = [t^{*(n)} - (t_r - \hat{t}_a) : t^{*(n)} + F],$$

where $\hat{t}_a$ is the predicted incident occurrence time from Eq.(4). It should be noted that $PW^{(n)}$ has the same length as PW .

Recall our problem statement in Section 5.3. It is of interest to predict $c(e_{p_i}^w, PW)$, but this is difficult because the speed data for PW are not available at time t_r . Therefore, we propose a method for predicting $c(e_{p_i}^w, PW)$ using historical speed data for $PW^{(n)}$, $n = 1, \dots, N$.

For each $e_{p_i}^w \in \{N_{in}^k(e_0)\}_{k=1}^K$, we can obtain $c(e_{p_i}^w, PW^{(n)})$, $n = 1, \dots, N$,

$$c(e_{p_i}^w, PW^{(n)}) = \begin{cases} 1, & \text{if } e_{p_i}^w \text{ is congested in } PW^{(n)} \\ 0, & \text{otherwise,} \end{cases}$$

according to Definition 5. $e_{p_i}^w$ is congested in $PW^{(n)}$ if $\exists t \in PW^{(n)}$ such that

$$\max s(e_{p_i}^w, [t - 10 : t]) < 0.7 \times ff(e_{p_i}^w),$$

where $ff(e_{p_i}^w)$ is the maximum speed restriction of $e_{p_i}^w$.

Once $c(e_{p_i}^w, PW^{(n)})$, $n = 1, \dots, N$, are obtained, we can predict $c(e_{p_i}^w, PW)$ as

$$\hat{c}(e_{p_i}^w, PW) = \begin{cases} 1, & \text{if } \frac{1}{N} \sum_{n=1}^N c(e_{p_i}^w, PW^{(n)}) > 0.5 \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

The procedure in Eq.(22) is repeated for $w = 1, \dots, W$, $i = 1, \dots, K$

For a path $P_w = (e_0 e_{p_1}^w e_{p_2}^w \dots e_{p_K}^w) \in P^K(e_0)$, $\hat{c}(e_{p_i}^w, PW)$, $i = 1, \dots, K$, yield the predicted sequence of propagation indicators, as defined in Eq.(14).

$$\widehat{p-path}(P_w, PW).$$

After obtaining the predicted sequences of propagation indicators for all paths in $P^K(e_0)$, we define a *propagation matrix* as

$$\widehat{PM} := \begin{bmatrix} \widehat{p-path}(P_1, PW) \\ \widehat{p-path}(P_2, PW) \\ \vdots \\ \widehat{p-path}(P_W, PW) \end{bmatrix}. \quad (23)$$

5.4 Experimental Results

The performance of the proposed method is evaluated using 215 real traffic incidents and related user trajectory data obtained from one of the major GPS navigation systems in South Korea. Experiments code is developed by Python 3.8 on AMD Ryzen Threadripper 3970X 32-Core processor workstation with Nvidia Titan RTX 24GB.

Dataset

We further filtered incidents data already filtered in Section 4.5, which are 215 incident cases without any changes in incoming roads topological location. This is to fairly apply the pattern matching technique to all roads with same amount of past duration.

The incident dataset used in this study is provided by the Korea National Police Agency and contained 215 incidents that occurred from September 2 to December 1 of 2020 in Seoul, Korea. The red circles in Fig. 3 represent the locations of the 215 traffic incidents considered in this study. Let $e_{0,u} \in E$,

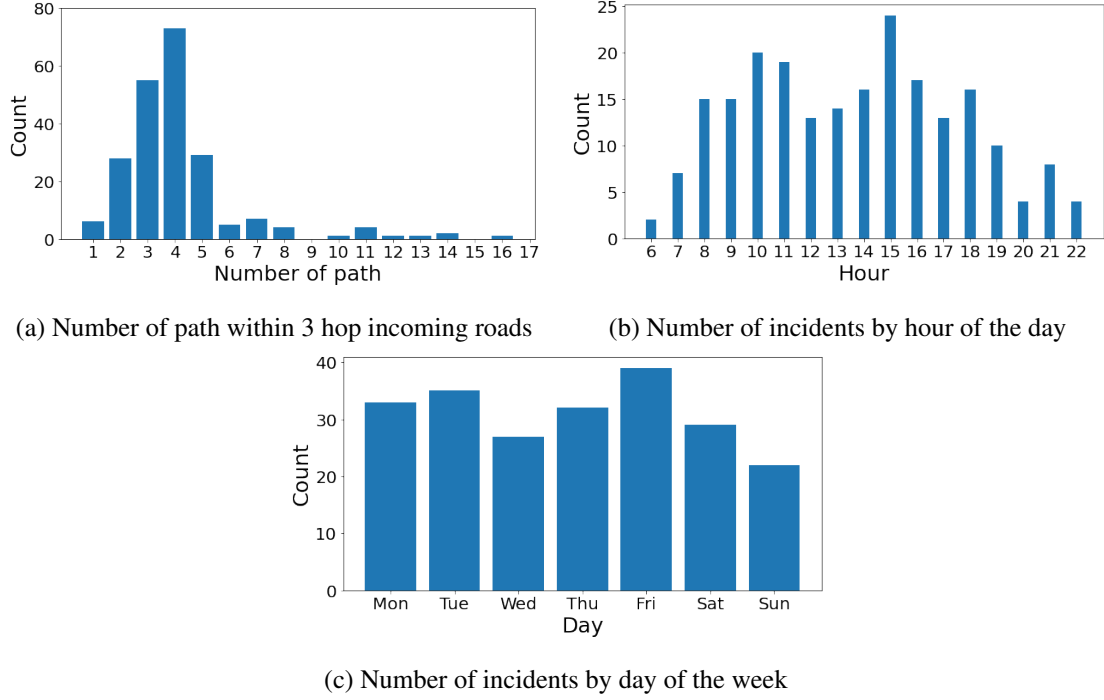


Figure 8: Summary of the traffic incident data

$u = 1, \dots, 215$, denote the incident roads, described by both geographical coordinates and a road segment ID. Fig. 8(a) illustrates the histogram of $P^3(e_{0,u})$, which is the number of paths within 3-hop incoming neighbors. Each $e_{0,u}$ is associated with a reported time $t_{r,u}$. Fig. 8(b) and Fig. 8(c) show the distribution of hour of the day and day of the week at the time of occurrence, $t_{r,u}$, respectively.

The trajectory dataset consists of a series of points with latitude, longitude, and timestamp information generated when vehicles travel with navigation systems. The user trajectory data is converted into average speed data, $\{s(e, t) \in S : E \times T \rightarrow \mathbb{R}^+\}$, for all road segment in Seoul, $e \in E$, over the given time, $t \in T$, through a map matching process [42].

Experimental Setup

Over the given time period, the speed data for all roads are normalized using their maximum speed restriction. For each traffic incident $e_{0,u}$, we applied the proposed method and predicted sequences of propagation indicators for all paths in $P^3(e_{0,u})$, $u = 1, \dots, 215$, and obtained the corresponding propagation matrix in Eq.(23).

The performance of the propagation prediction is evaluated by accuracy, recall, and precision. Under our experimental setup, accuracy is defined as

$$Accuracy = \frac{1}{215} \sum_{u=1}^{215} ACC_u, \quad (24)$$

where

$$ACC_u = \frac{\sum_{i=1}^3 \sum_{e_j \in N_{in}^i(e_{0,u})} \mathbb{1}(q(e_j, PW) = \hat{q}(e_j, PW))}{\sum_{i=1}^3 |N_{in}^i(e_{0,u})|}.$$

Recall and precision are defined in a similar way. Using these classification evaluation metrics, Section 5.4 carried out a comparison study with the existing benchmarks and Section 5.4 applied the proposed propagation prediction in real-time scenarios.

Hyper-parameter Optimization

The proposed method includes user-specific hyper-parameters such as N in Eq.(19), PT in Eq.(20), and F in Eq.(21). For hyper-parameter optimization, we performed a grid search in the following hyper-parameter search space:

$$N = \{1, \dots, 30\}, PT = \{0, 30\}, F = \{60, 120, 180\}.$$

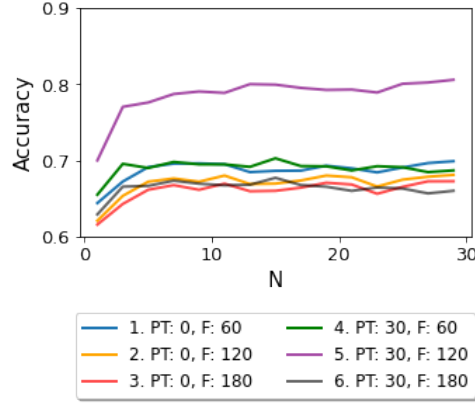
Table 4: Results of a grid search in the hyper-parameter search space

	$PT = 0$			$PT = 30$		
	$F = 60$	120	180	$F = 60$	120	180
$N \leq 10$	0.696	0.677	0.668	0.698	0.790	0.674
$N \leq 20$	0.696	0.680	0.671	0.703	0.800	0.677
$N \leq 30$	0.699	0.681	0.673	0.703	0.806	0.677

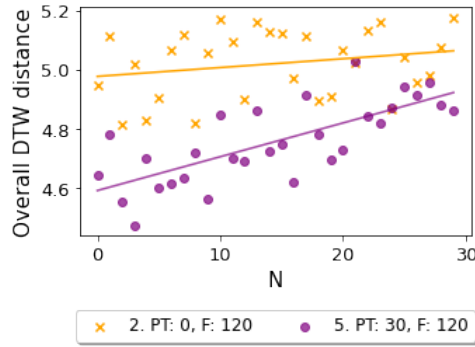
Table 4 and Fig. 5.4(a) show the results of hyper-parameter optimization based on accuracy in Eq.(24). As a result, the combination of $PT = 30$ and $F = 120$ outperforms the others, which is the purple solid line in the figures. For the combination of $PT = 30$ and $F = 120$, we observed that the value of accuracy increases rapidly until $N = 5$ and then increases gradually after that. Hence, we can stress that 10 ~ 20 similar historical source traffic patterns are sufficient for predicting future congestion propagation patterns while maintaining computational efficiency and robustness.

Recall that PT represents the length of the free-flow phase in the searching window (SW). By adding traffic patterns in the free-flow phase to the SW , we expect to be able to detect more similar source traffic patterns than when they are not added. To confirm the effect of adding PT , we compared the average DTW distances of 215 incidents between $PT = 0$ and $PT = 30$.

In order to verify this finding, we measured DTW distance between $s(e_0, t_r - 120 : t_r + 180)$ and $(e_0, t^{*(n)} - 120 : t^{*(n)} + 180)$ for $n = 1, \dots, N$. We set such time range to reflect the overall traffic speed time series from ‘free-flow phase’ to even after t_r for incident road e_0 . Fig. 5.4(b) shows averaged overall DTW distance by all 215 incident cases. It shows that every DTW distances calculated with PT being equal to 30 min are lower than DTW distance without PT for all different SW conditions. In other words, the historical speed time series of incident road e_0 selected according to the size of SW with PT being 30 min are overall similar to speed time series of real incident case. Consequently, the average DTW distances with $PT = 30$ outperformed the one with $PT = 0$, which means that the SW with the free-flow phase is effective to detect similar source traffic patterns.



(a) Relationship between N and Accuracy



(b) Overall DTW distance comparison for incident road

Figure 9: Results of hyper-parameter optimization

Model Comparison

The performance of the proposed method is compared with comparable benchmarking methods using 215 real traffic incidents and related user trajectory data. The hyper-parameters are set at $PT = 30$, $F = 120$, and $N = 13$.

We considered three existing benchmarks: traffic flow pattern classification using K-means clustering [74], short-term traffic forecasting using multivariate autoregressive models [75], and speed prediction using Graph WaveNet [76].

K-means algorithm is proposed to classify traffic speed pattern [74]. We divide historical speed into $s(e_0, [t - L : t])$ for any t before accident reported. Each time series is an independent instance and we generate K clusters with the historical data. In case of the current situation, $s(e_0, [t_r - L : t_r])$ is clustered into cluster k^* using pre-trained model. Therefore predicted pattern should be similar to speed patterns in the cluster k^* from historical data. To avoid locally similar patterns, we also implemented a grouping search algorithm in Algorithm 1. We use mini-batch k-means with $K = 10$ for computational efficiency with using scikit-learn¹.

VAR model, multivariate time series prediction method, is proposed to predict short-term traffic

¹<https://github.com/scikit-learn/scikit-learn>

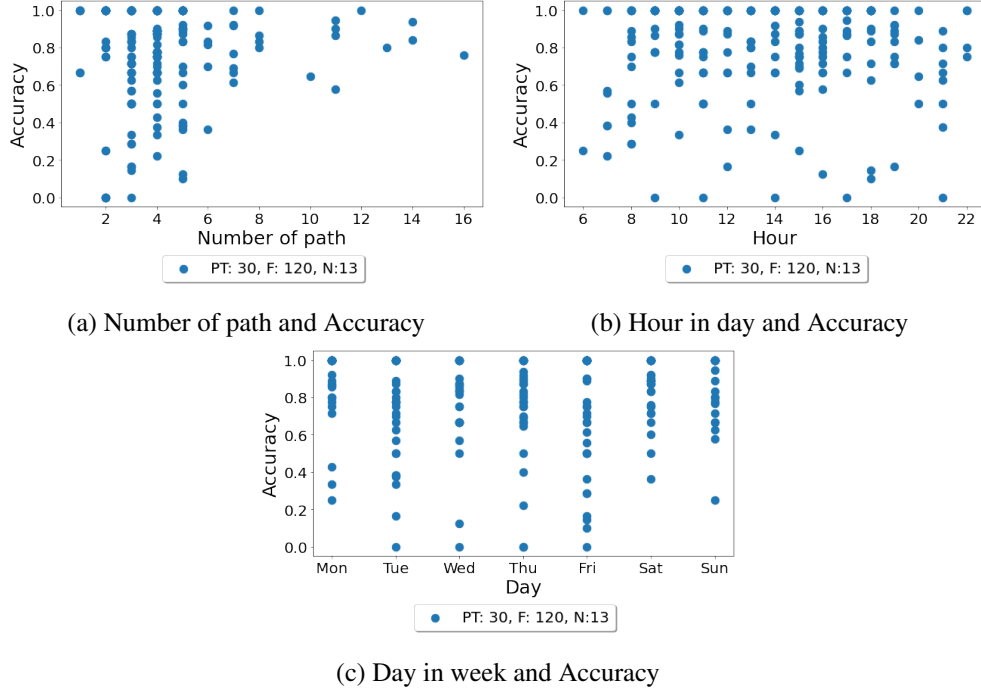


Figure 10: Accuracy distribution

speed in [75]. In our case, VAR model is fitted by prior 90 days speed distribution to predict speeds. We are interested in modeling $T \times N_{in}^k(e_0)$ multivariate time series Y , where T denotes the number of observations and $N_{in}^k(e_0)$ the number of roads in the network. One way of estimating relationships between the time series and their lagged values is the vector autoregression process:

$$Y_t = v + A_1 Y_{t-1} + \dots + A_p Y_{t-p} + u_t \quad (25)$$

$$u_t \sim \text{Normal}(0, \Sigma_u)$$

where A_i is $N_{in}^k(e_0) \times N_{in}^k(e_0)$ coefficient matrix. VAR algorithm in our experiment is implemented by statmodel².

Graph WaveNet is one of the SOTA method in the field of traffic speed time series prediction. This model uses graph convolution network (GCN) with a self-adaptive adjacency matrix to automatically learn latent static graph structures of traffic network [76]. Given a traffic network, we set the model to learn historical 120 min of speed and predict next 120 min speed time series. The model was implemented with epoch of 10, batch size of 64, and a learning rate of 0.001. The model is implemented by pytorch version³.

In summary, we use K-means algorithms instead of DTW as a method for pattern matching when generating $\widehat{PM}$, and design VAR and Graph Wavenet to generate $\widehat{PM}$ based on the predicted speed time series in each model.

We compare the model performance of prediction of propagation matrix with accuracy, recall, precision and computation time. As in all information retrieval, the result compromises between precision and

²<https://github.com/statsmodels/statsmodels>

³<https://github.com/nnzhan/Graph-WaveNet>

recall, minimizing false positives and misses. In practice, the model should satisfy to maximize recall subject to that precision constraint. Since our interest is to predict how congestion propagation spread in real traffic incident situation, we use recall value which quantifies the number of correct congestion propagation predicted for true propagation patterns.

Table 5: Performance comparison

Models	Accuracy	Recall	Precision	Time(s)/ Incident
DTW	0.800	0.813	0.895	23.955
K-means	0.563	0.501	0.891	24.247
VAR	0.755	0.688	0.951	13.270
Graph Wavenet	0.714	0.611	0.768	102.474

As shown in Table 5, the proposed pattern matching method outperformed all three benchmarks in terms of accuracy and recall values. In the case of K-means, it shows extremely low performance in terms of accuracy and recall. This suggests that matching incident time series patterns, in particular where dramatic speed curves occur, with Euclidean distance is not significant. The traffic forecasting method and the speed prediction method performed better than the pattern classification method in general, but they were poor in recall. It should be noted that recall values are more important in the context of congestion propagation prediction. The poor recall values resulted from that the existing methods failed to learn information of non-recurrent traffic congestion effectively.

The performance of VAR and Graph Wavenet are similar to our performance in terms of accuracy, but the recall value, which is actually more important in real-life situations, seems to drop significantly. This is because they are trained based on speed data over the past 90 days, which have stationarity, and did not learn information about non-recurrent congestion pattern such as incident situations. That is why they tend to predict speed of $N_{in}^k(e_0)$ stationary, which decreases the recall value.

Real-time Experiments

We applied the proposed method in real-time scenarios. Suppose that the latest speed data comes after t_r and we want to update the proposed model with the new data. Even for real-time scenarios, the proposed method is suitable because only the searching window in Eq.(20) needs to be modified as follows.

$$SW = [\hat{t}_a - PT : t_r + RU]. \quad (26)$$

Table 6 showed the performance of the proposed model with various values of RU in Eq.(26). As expected, the accuracy values increased in the following order: 0.792, 0.801, and 0.811 as the value of RU increased. Similarly, the recall values increased in the order of 0.802, 0.828, and 0.836. This was because the broader SW reflected more information about non-recurrent traffic patterns on the incident roads. In the process of matching the target traffic pattern with the historical source traffic patterns,

Table 6: Prediction results in real-time scenarios with $PT = 30$, $F = 120$, $N = 13$

Metric	$RU = 0$	$RU = 10$	$RU = 20$
Accuracy	0.800	0.803	0.816
Recall	0.813	0.826	0.840
Preicion	0.895	0.889	0.882

the more similar source traffic patterns were selected, which resulted in the more accurate propagation prediction.

VI Conclusion

6.1 Summary and Contributions

Providing of various aspects of incident impact and prior knowledge of critical factors contributing each aspect are important for both traffic operators and any corporation that provide navigation services in order to reduce the influence of an incident and to provide drivers with alternative route in real-time manner. In this research, under the limited situation where only access to incident related information of incident report time and the speed is guaranteed we proposed a framework to quantify incident impact in spatial, temporal, and speed aspect and acquired reasonable feature importance contributing each aspect by utilizing the method called LIME (Local Interpretable Model-agnostic Explanations) on real traffic incident cases in Seoul, South Korea.

We presented a novel online learning method based on pattern matching to predict the propagation patterns of non-recurrent traffic congestion. Also, we developed a grouping search algorithm for effective pattern matching. By using pattern matching techniques, our model effectively matched similar historical speed pattern of an incident road and predicted the future non-recurrent congestion propagation patterns with high accuracy and recall values. This results demonstrated the hypothesis that the current non-recurrent propagation pattern evolves similarly to the historical recurrent pattern. Furthermore, we validated the applicability to real-time scenarios by improving the performance of the model as the relevant information is updated in real-time.

Unlike traditional machine learning methods or deep learning methods that are focused on recurrent patterns and require pre-defined traffic network and corresponding parameters, the proposed method can be applied to any incident anytime, anywhere. This opens up a new direction in modelling non-recurrent congestion propagation patterns.

6.2 Limitation and Future research

However, there are some limitations regarding the data used in this study. Those limitations are described as follows. First, data might contains noisy data caused by each a driver and hard to be defined the purpose for each trajectory route [77, 78]. Second, data collected from the Navigation is user based, such that it might not be representative of the traffic conditions for every road segments [78]. In case of narrow alleys, there are quite many NaN values since the number of trajectory data from users go through such roads are not enough and unreliable. Third, there are absence of independent variables that explain the uniqueness of an traffic incident, such as ‘number of casualty or injured’, ‘number of vehicle involved’, or ‘number of lane blocked’. These are the importance independent variables that distinguish each traffic incident and have close relationship with the incident impact resulted [46].

If more detailed incident data and precise speed data are obtained, more accurate prediction on incident impacts and more precise interpretation of the related feature importance can be obtained for making delicate traffic plans to mitigate the adverse effect of a traffic incident. Although there are some restrictions regarding data used, we proposed the new frame to analyze the incident impact based on

speed data and incident reported time. The new metrics include not only the duration of the incident but also the spatial and speed dynamic aspects through propagation level and speed drop ratio. Therefore, the method can not only be applied to any road network where incident data, including speed, are collected, but can also be used in real-world situations where incident-related information is only ‘incident report time’. Using the proposed method, transport policy and plan level analysis related to the design of transport systems or preparation of operational plans for safety can be evaluated and optimal routes can be proposed to navigation users in the event of an incident.

In future work, we will try to match patterns using other traffic indicators such as volume, flow, and density that can explain the accident situation and explore approaches to learn dynamic nature of non-recurrent congestion propagation patterns.

References

- [1] R. W. Hall, “Non-recurrent congestion: How big is the problem? are traveler information systems the solution?” *Transportation Research Part C: Emerging Technologies*, vol. 1, no. 1, pp. 89–103, 1993.
- [2] A. Haghani, D. Iliescu, M. Hamed, and S. Yong, “Methodology for quantifying the cost effectiveness of freeway service patrols programs—case study: Help program, i-95 corridor coalition,” *Univ. of Maryland, College Park, MD*, 2006.
- [3] D. H. Kim, K. Y. Hwang, and Y. Yoon, “Prediction of traffic congestion in seoul by deep neural network,” *The Journal of The Korea Institute of Intelligent Transport Systems*, vol. 18, no. 4, pp. 44–57, 2019.
- [4] Y.-J. Lee and O. Min, “Long short-term memory recurrent neural network for urban traffic prediction: A case study of seoul,” in *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2018, pp. 1279–1284.
- [5] N. P. Agency, “Accident police-received,” https://www.police.go.kr/www/open/public2/public2018_04.jsp, May 2013.
- [6] F. Klinker, “Exponential moving average versus moving exponential average,” *Mathematische Semesterberichte*, vol. 58, no. 1, pp. 97–107, 2011.
- [7] M. Ausloos and K. Ivanova, “Mechanistic approach to generalized technical analysis of share prices and stock market indices,” *The European Physical Journal B-Condensed Matter and Complex Systems*, vol. 27, no. 2, pp. 177–187, 2002.
- [8] N. A. Harish, S. Mohd Hairullah, S. Abu Bakar, and K. Abd Rahman, “Fuzzy golden cross and fuzzy death cross as stock market forecasting indicator,” *ASM Science Journal*, vol. 12, no. Specia, pp. 120–124, 2019.
- [9] V. Hurdle, M. I. Merlo, and D. Robertson, “Study of speed-flow relationships on individual freeway lanes,” *Transportation Research Record*, vol. 1591, no. 1, pp. 7–13, 1997.
- [10] I. Ghosh, P. T. Savolainen, and T. J. Gates, “Examination of factors affecting freeway incident clearance times: a comparison of the generalized f model and several alternative nested models,” *Journal of Advanced Transportation*, vol. 48, no. 6, pp. 471–485, 2014.

- [11] C.-S. Li and M.-C. Chen, "A data mining based approach for travel time prediction in freeway with non-recurrent congestion," *Neurocomputing*, vol. 133, pp. 74–83, 2014.
- [12] R. Li, F. C. Pereira, and M. E. Ben-Akiva, "Overview of traffic incident duration analysis and prediction," *European transport research review*, vol. 10, no. 2, pp. 1–13, 2018.
- [13] M. Saarela and S. Jauhiainen, "Comparison of feature importance measures as explanations for classification models," *SN Applied Sciences*, vol. 3, no. 2, pp. 1–12, 2021.
- [14] H. C. Manual, "Trb special report 209," *National Research Council, Washington, DC*, 1994.
- [15] M. E. Goolsby, "Influence of incidents on freeway quality of service," *Highway Research Record*, no. 349, 1971.
- [16] Y. Chung and W. W. Recker, "A methodological approach for estimating temporal and spatial extent of delays caused by freeway accidents," *IEEE Transactions on Intelligent Transportation Systems*, vol. 13, no. 3, pp. 1454–1461, 2012.
- [17] S. C. Wirasinghe, "Determination of traffic delays from shock-wave analysis," *Transportation Research*, vol. 12, no. 5, pp. 343–348, 1978.
- [18] W.-M. Chow, "A study of traffic performance models under an incident condition," *Transportation research record*, vol. 567, pp. 31–36, 1976.
- [19] H. Al-Deek, A. Garib, and A. Radwan, "New method for estimating freeway incident congestion," *Transportation Research Record*, pp. 30–39, 1995.
- [20] F. He, X. Yan, Y. Liu, and L. Ma, "A traffic congestion assessment method for urban road networks based on speed performance index," *Procedia engineering*, vol. 137, pp. 425–433, 2016.
- [21] J. Shunping, P. Hongqin, and L. Shuang, "Urban traffic state estimation considering resident travel characteristics and road network capacity," *Journal of Transportation Systems Engineering and Information Technology*, vol. 11, no. 5, pp. 81–85, 2011.
- [22] "Documentation and definitions—urban congestion reports—operations performance measurement—fhwa operations." https://ops.fhwa.dot.gov/perf_measurement/ucr/documentation.htm, accessed: 2021-02-04.
- [23] M. G. Karlaftis and E. I. Vlahogianni, "Statistical methods versus neural networks in transportation research: Differences, similarities and some insights," *Transportation Research Part C: Emerging Technologies*, vol. 19, no. 3, pp. 387–399, 2011.
- [24] A. Garib, A. Radwan, and H. Al-Deek, "Estimating magnitude and duration of incident delays," *Journal of Transportation Engineering*, vol. 123, no. 6, pp. 459–466, 1997.
- [25] S. Peeta, J. L. Ramos, and S. Gedela, "Providing real-time traffic advisory and route guidance to manage borman incidents on-line using the hoosier helper program," 2000.

- [26] A. J. Khattak, J. L. Schofer, and M.-H. Wang, “A simple time sequential procedure for predicting freeway incident duration: Ivhs journal,” 1994.
- [27] Y. Chung, “Development of an accident duration prediction model on the korean freeway systems,” *Accident Analysis & Prevention*, vol. 42, no. 1, pp. 282–289, 2010.
- [28] A. T. Hojati, L. Ferreira, S. Washington, and P. Charles, “Hazard based models for freeway traffic incident duration,” *Accident Analysis & Prevention*, vol. 52, pp. 171–181, 2013.
- [29] Y. Chung and B.-J. Yoon, “Analytical method to estimate accident duration using archived speed profile and its statistical analysis,” *KSCE Journal of Civil Engineering*, vol. 16, no. 6, pp. 1064–1070, 2012.
- [30] N. E. Breslow, “Analysis of survival data under the proportional hazards model,” *International Statistical Review/Revue Internationale de Statistique*, pp. 45–57, 1975.
- [31] D. Nam and F. Mannering, “An exploratory hazard-based analysis of highway incident duration,” *Transportation Research Part A: Policy and Practice*, vol. 34, no. 2, pp. 85–102, 2000.
- [32] D. S. Bennett, “Parametric models, duration dependence, and time-varying data revisited,” *American Journal of Political Science*, pp. 256–270, 1999.
- [33] J.-T. Lee and J. Fazio, “Influential factors in freeway crash response and clearance times by emergency management services in peak periods,” *Traffic injury prevention*, vol. 6, no. 4, pp. 331–339, 2005.
- [34] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv preprint arXiv:1702.08608*, 2017.
- [35] R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, and D. Pedreschi, “A survey of methods for explaining black box models,” *ACM computing surveys (CSUR)*, vol. 51, no. 5, pp. 1–42, 2018.
- [36] C. Molnar, *Interpretable machine learning*. Lulu. com, 2020.
- [37] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why should i trust you?: Explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. ACM, 2016, pp. 1135–1144.
- [38] S. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *arXiv preprint arXiv:1705.07874*, 2017.
- [39] Z. C. Lipton, “The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery,” *Queue*, vol. 16, no. 3, pp. 31–57, 2018.
- [40] T. Afrin and N. Yodo, “A survey of road traffic congestion measures towards a sustainable and resilient transportation system,” *Sustainability*, vol. 12, no. 11, p. 4660, 2020.

- [41] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [42] P. Newson and J. Krumm, "Hidden markov map matching through noise and sparseness," in *Proceedings of the 17th ACM SIGSPATIAL international conference on advances in geographic information systems*, 2009, pp. 336–343.
- [43] L. C. Freeman, "Centrality in social networks conceptual clarification," *Social networks*, vol. 1, no. 3, pp. 215–239, 1978.
- [44] J. Zhang and Y. Luo, "Degree centrality, betweenness centrality, and closeness centrality in social network," in *2017 2nd International Conference on Modelling, Simulation and Applied Mathematics (MSAM2017)*. Atlantis Press, 2017, pp. 300–303.
- [45] A. Saxena, V. Malik, and S. Iyengar, "Estimating the degree centrality ranking," in *2016 8th International Conference on Communication Systems and Networks (COMSNETS)*. IEEE, 2016, pp. 1–2.
- [46] H. Park, A. Haghani, and X. Zhang, "Interpretation of bayesian neural networks for predicting the duration of detected incidents," *Journal of Intelligent Transportation Systems*, vol. 20, no. 4, pp. 385–400, 2016.
- [47] B. Š. Motnikar, T. Pisanski, and D. Čepar, "Time-series forecasting by pattern imitation," *Operations-Research-Spektrum*, vol. 18, no. 1, pp. 43–49, 1996.
- [48] H. Xiong, A. Vahedian, X. Zhou, Y. Li, and J. Luo, "Predicting traffic congestion propagation patterns: a propagation graph approach," in *Proceedings of the 11th ACM SIGSPATIAL International Workshop on Computational Transportation Science*, 2018, pp. 60–69.
- [49] S. Basak, A. Dubey, and L. Bruno, "Analyzing the cascading effect of traffic congestion using lstm networks," in *2019 IEEE International Conference on Big Data (Big Data)*. IEEE, 2019, pp. 2144–2153.
- [50] N. Kumar and M. Raubal, "Applications of deep learning in traffic congestion alleviation: A survey," *arXiv preprint arXiv:2102.09759*, 2021.
- [51] W. Liu, Y. Zheng, S. Chawla, J. Yuan, and X. Xing, "Discovering spatio-temporal causal interactions in traffic data streams," in *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2011, pp. 1010–1018.
- [52] Z. Wang, M. Lu, X. Yuan, J. Zhang, and H. Van De Wetering, "Visual traffic jam analysis based on trajectory data," *IEEE transactions on visualization and computer graphics*, vol. 19, no. 12, pp. 2159–2168, 2013.
- [53] S. Sun, J. Chen, and J. Sun, "Traffic congestion prediction based on gps trajectory data," *International Journal of Distributed Sensor Networks*, vol. 15, no. 5, p. 1550147719847440, 2019.

- [54] A. Zhang and Z. Gao, "Effect of atis information under incident-based congestion propagation," *Procedia-Social and Behavioral Sciences*, vol. 43, pp. 628–637, 2012.
- [55] Z. Liang and Y. Wakahara, "Real-time urban traffic amount prediction models for dynamic route guidance systems," *EURASIP Journal on Wireless Communications and Networking*, vol. 2014, no. 1, pp. 1–13, 2014.
- [56] W.-p. Fei, G.-h. Song, F. Zhang, Y. Gao, and L. Yu, "Practical approach to determining traffic congestion propagation boundary due to traffic incidents," *Journal of Central South University*, vol. 24, no. 2, pp. 413–422, 2017.
- [57] V. Fabrizi and R. Ragona, "A pattern matching approach to speed forecasting of traffic networks," *European Transport Research Review*, vol. 6, no. 3, pp. 333–342, 2014.
- [58] R. Adhikari and R. K. Agrawal, "An introductory study on time series modeling and forecasting," *arXiv preprint arXiv:1302.6613*, 2013.
- [59] A. Dimri, H. Singh, N. Aggarwal, B. Raman, K. Ramakrishnan, and D. Bansal, "Barosense: Using barometer for road traffic congestion detection and path estimation with crowdsourcing," *ACM Transactions on Sensor Networks (TOSN)*, vol. 16, no. 1, pp. 1–24, 2019.
- [60] J. Wu, Z. Cui, V. S. Sheng, Y. Shi, and P. Zhao, "Mixed pattern matching-based traffic abnormal behavior recognition," *The Scientific World Journal*, vol. 2014, 2014.
- [61] R. Asadi and A. Regan, "Spatio-temporal clustering of traffic data with deep embedded clustering," in *Proceedings of the 3rd ACM SIGSPATIAL International Workshop on Prediction of Human Mobility*, 2019, pp. 45–52.
- [62] E. J. Horvitz, J. Apacible, R. Sarin, and L. Liao, "Prediction, expectation, and surprise: Methods, designs, and study of a deployed traffic forecasting service," *arXiv preprint arXiv:1207.1352*, 2012.
- [63] I. Lana, J. Del Ser, M. Velez, and E. I. Vlahogianni, "Road traffic forecasting: Recent advances and new challenges," *IEEE Intelligent Transportation Systems Magazine*, vol. 10, no. 2, pp. 93–109, 2018.
- [64] E. L. Manibardo, I. Laña, J. L. Lobo, and J. Del Ser, "New perspectives on the use of online learning for congestion level prediction over traffic data," in *2020 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2020, pp. 1–8.
- [65] D. Laney *et al.*, "3d data management: Controlling data volume, velocity and variety," *META group research note*, vol. 6, no. 70, p. 1, 2001.
- [66] A. Bifet, R. Gavaldà, G. Holmes, and B. Pfahringer, *Machine learning for data streams: with practical examples in MOA*. MIT press, 2018.

- [67] S. Kamran and O. Haas, "A multilevel traffic incidents detection approach: identifying traffic patterns and vehicle behaviours using real-time gps data," in *2007 IEEE Intelligent Vehicles Symposium*. IEEE, 2007, pp. 912–917.
- [68] Z. Wei, C. Ande, and J. Guiyan, "Sampling and transmitting intervals optimization based on gps equipped floating car," in *2009 Second International Conference on Intelligent Computation Technology and Automation*, vol. 4. IEEE, 2009, pp. 97–100.
- [69] R. Dhakad and M. Jain, "Gps based road traffic congestion reporting system," in *2014 IEEE International Conference on Computational Intelligence and Computing Research*. IEEE, 2014, pp. 1–6.
- [70] E. D'Andrea and F. Marcelloni, "Detection of traffic congestion and incidents from gps trace analysis," *Expert Systems with Applications*, vol. 73, pp. 43–56, 2017.
- [71] Z. Bankó and J. Abonyi, "Correlation based dynamic time warping of multivariate time series," *Expert Systems with Applications*, vol. 39, no. 17, pp. 12 814–12 823, 2012.
- [72] B. S. Kerner and S. L. Klenov, "A microscopic model for phase transitions in traffic flow," *Journal of Physics A: Mathematical and General*, vol. 35, no. 3, p. L31, 2002.
- [73] J. Wang, W. Xie, B. Liu, D. R. Ragland *et al.*, "Identification of freeway secondary accidents with traffic shock wave detected by loop detectors," *Safety science*, vol. 87, pp. 195–201, 2016.
- [74] M. A. Silgu, H. B. Çelikoğlu *et al.*, "K-means clustering method to classify freeway traffic flow patterns," *Pamukkale University Journal of Engineering Sciences*, vol. 20, no. 6, pp. 232–239, 2014.
- [75] D. Pavlyuk, "Short-term traffic forecasting using multivariate autoregressive models," *Procedia Engineering*, vol. 178, pp. 57–66, 2017.
- [76] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph wavenet for deep spatial-temporal graph modeling," *arXiv preprint arXiv:1906.00121*, 2019.
- [77] J. Choi, K. Lee, H. Kim, S. An, and D. Nam, "Classification of inter-urban highway drivers' resting behavior for advanced driver-assistance system technologies using vehicle trajectory data from car navigation systems," *Sustainability*, vol. 12, no. 15, p. 5936, 2020.
- [78] Z. Kan, L. Tang, M.-P. Kwan, C. Ren, D. Liu, and Q. Li, "Traffic congestion analysis at the turn level using taxis' gps trajectory data," *Computers, Environment and Urban Systems*, vol. 74, pp. 229–243, 2019.

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